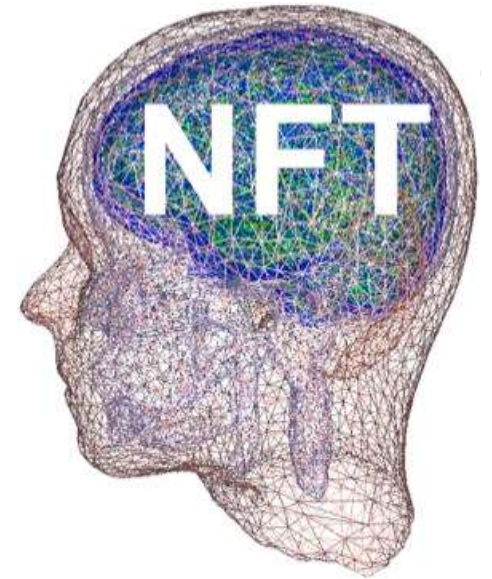




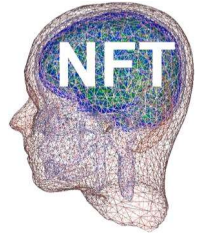
NFT

Neuroelectromagnetic Forward and Inverse Head Modeling Toolbox

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Swartz Center for Computational Neuroscience

Virtual EEGLAB workshop, June 2021



NFT code

<https://github.com/sccn/NFT>

Demo folders:

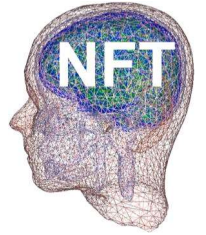
NFTplugin_demo_dipole

NFTplugin_demo_cortical

<https://rdl-share.ucsd.edu/message/U7F0uMivgbGZJur64TXDac>

zeynep@sccn.ucsd.edu

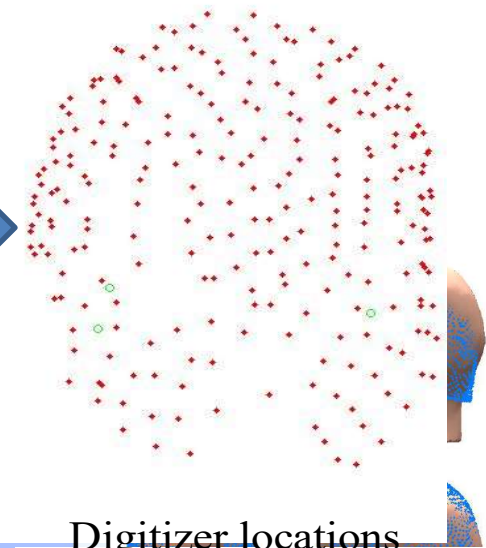
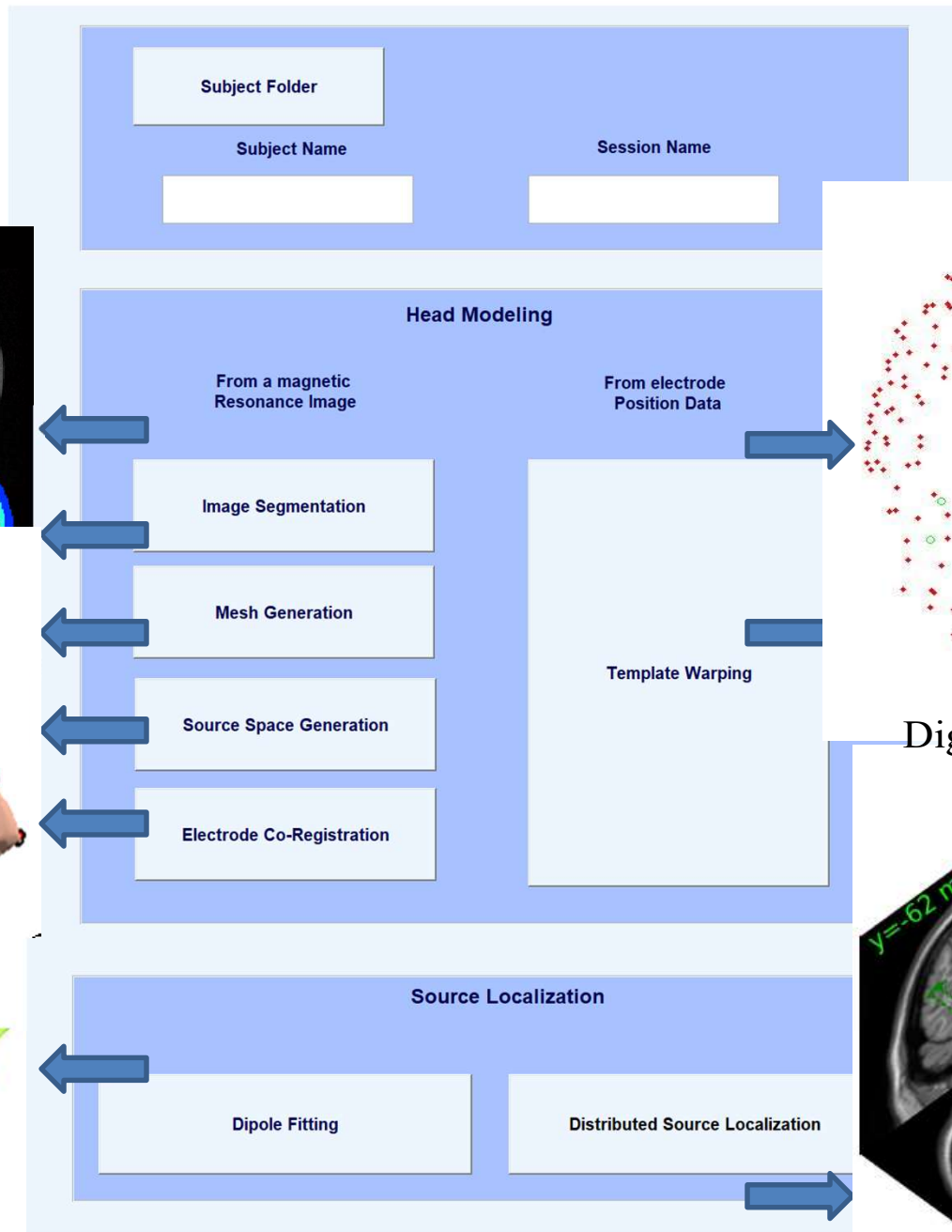
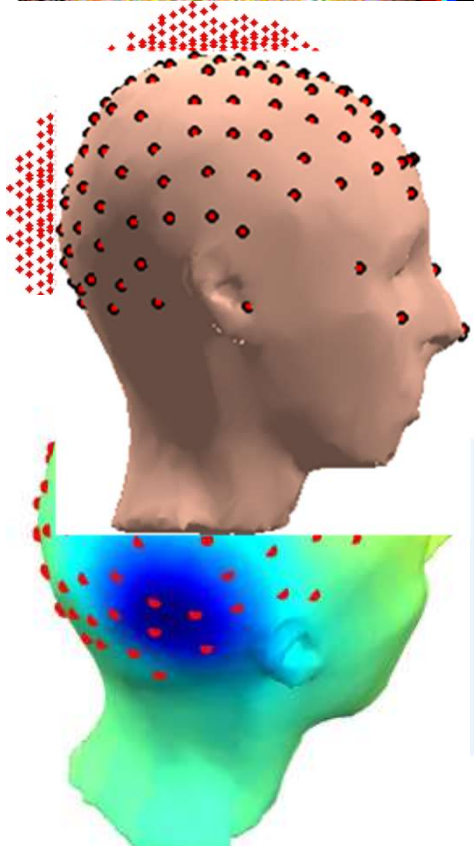
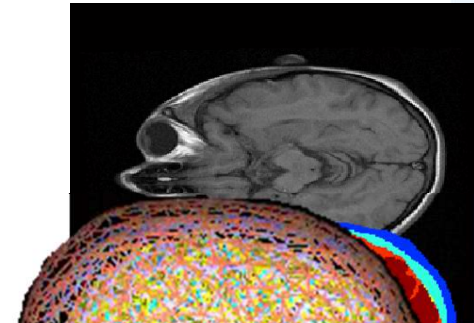
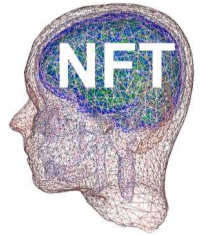
NFT



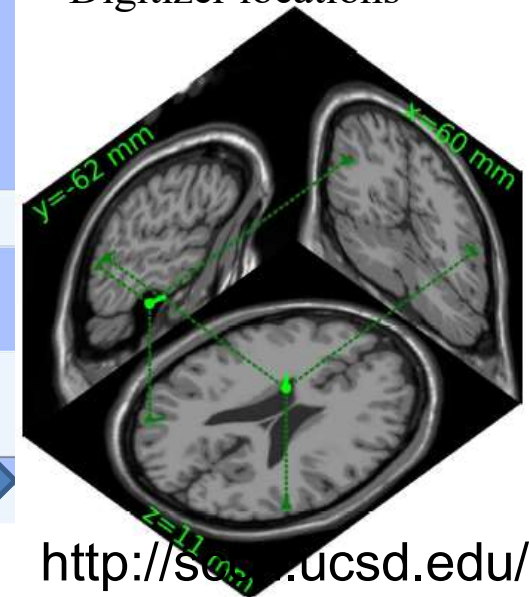
- ◆ A complete framework for accurate forward and inverse problem solution.
- ◆ Easy-to-use MATLAB environment with GUI and command-line functions.
- ◆ Ability to use available subject information
 - T1-weighted 3D MR images
 - Digitized sensor (electrode) locations

Comparison with Dipfit

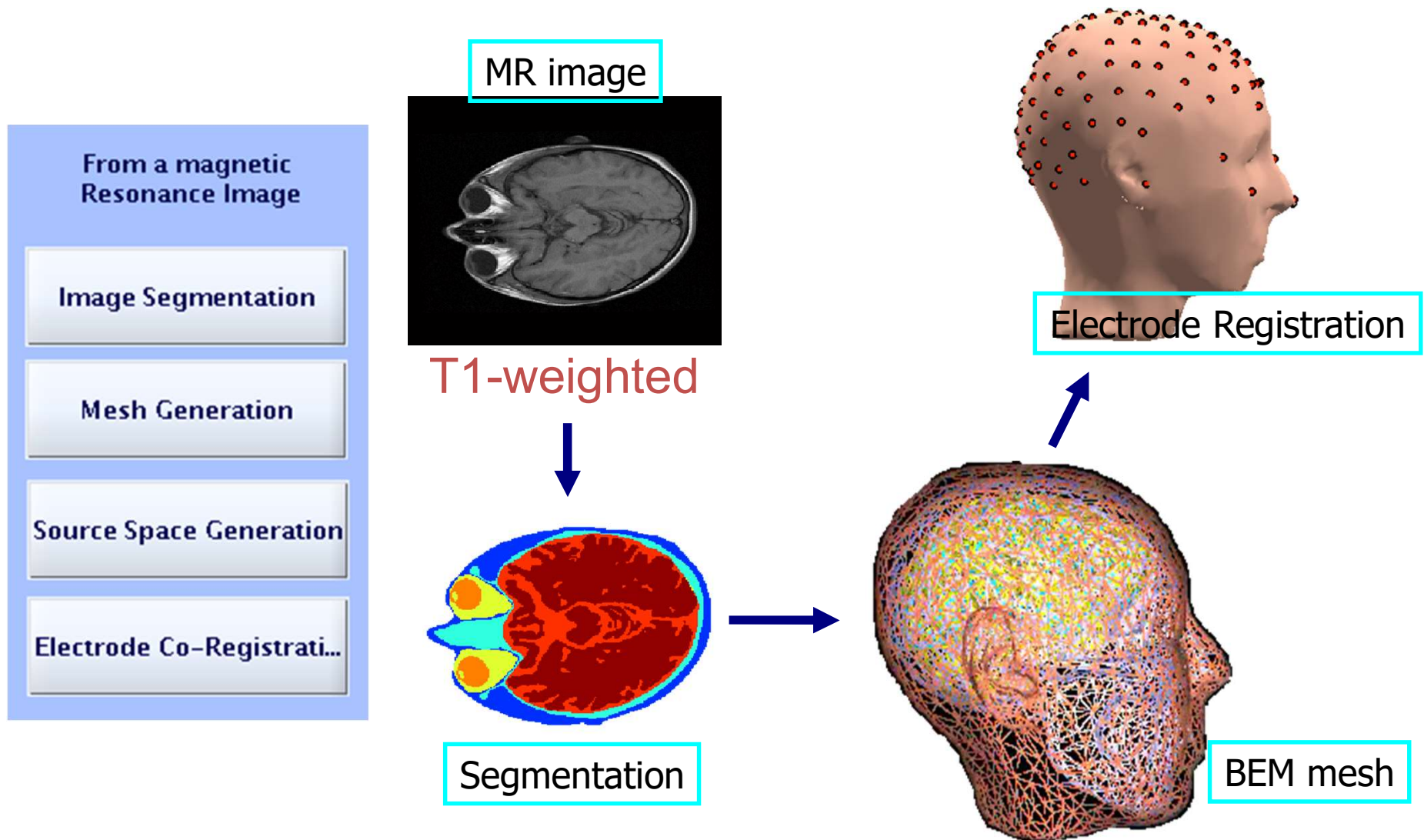
- ◆ The realistic model in Dipfit is a three-layer MNI head model represented with 3000 vertices.
 - The forward matrices are pre-calculated, so there is no need for FP calculations. Faster solutions.
- ◆ NFT generates subject-specific models.
 - NFT does model generation and forward problem calculations.
 - More accurate.



Digitizer locations



Head modeling from MR images



Preparing the MR Image

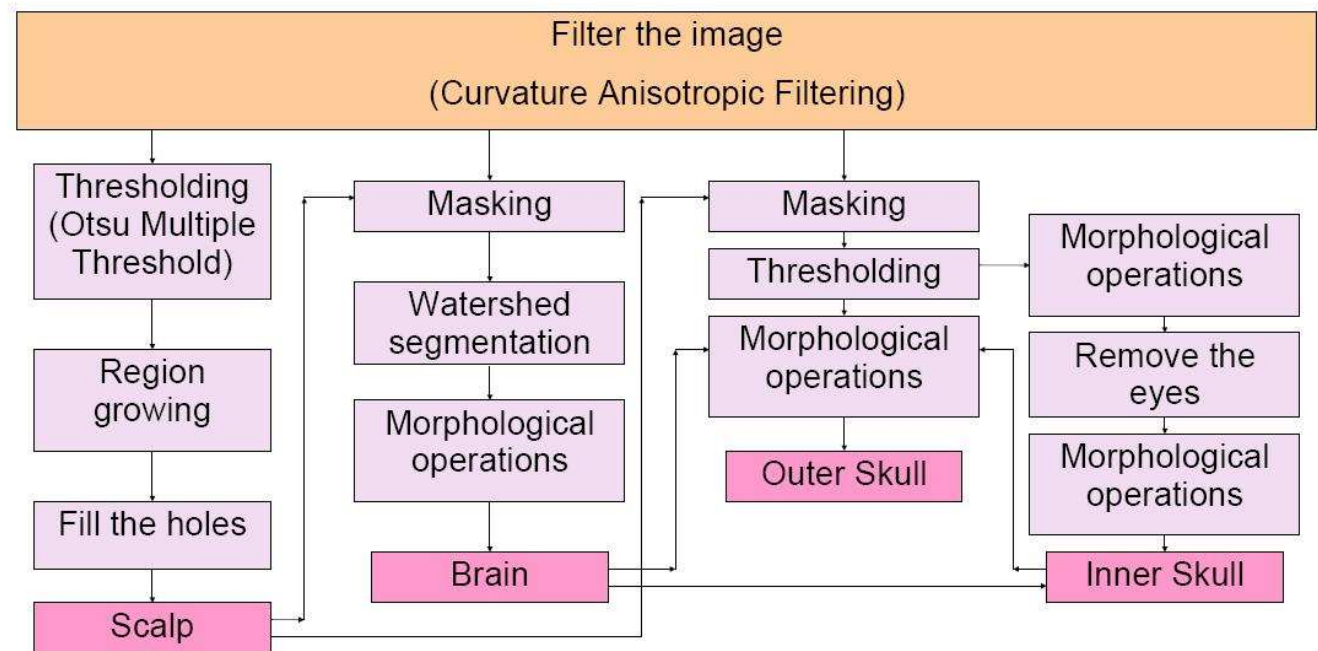
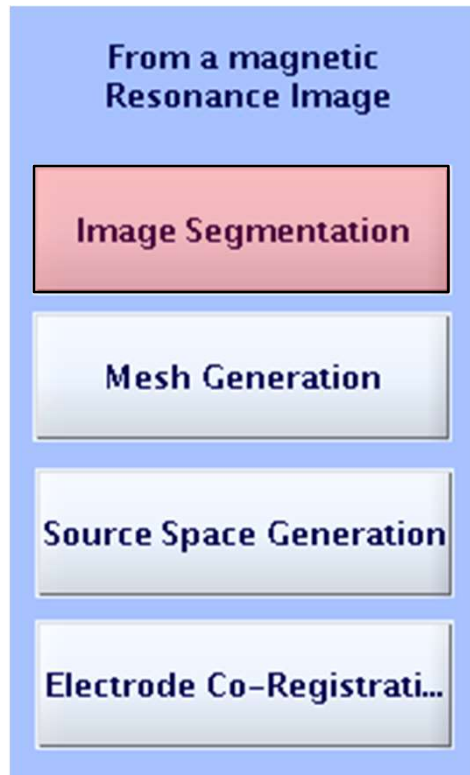
- ◆ Using FreeSurfer
 - Inhomogeneity correction
 - Convert to 1x1x1 volume
 - Arrange direction of the image
 - Save in analyze format

```
mri_convert -i brain.nii -ot analyze -o test1.img
```

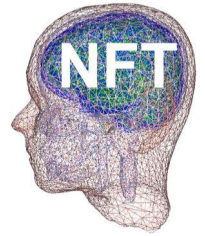
```
mri_nu_correct.mni --i test1.img --o t1c.img --n 2
```

```
mri_convert -i t1c.img --conform_size 1 --  
out_orientation PSR -ot analyze -o test2.img
```

Image Segmentation



Classifies four tissues from T1-weighted images
Scalp, Skull, CSF and Brain



Starting NFT

- ◆ To start from EEGLAB

EEGLAB -> Tools -> NFT

- ◆ To start as a standalone toolbox

addpath NFT directory

Type 'NFT' in Matlab

Demo folders: NFTplugin_demo_dipole

NFTplugin_demo_cortical

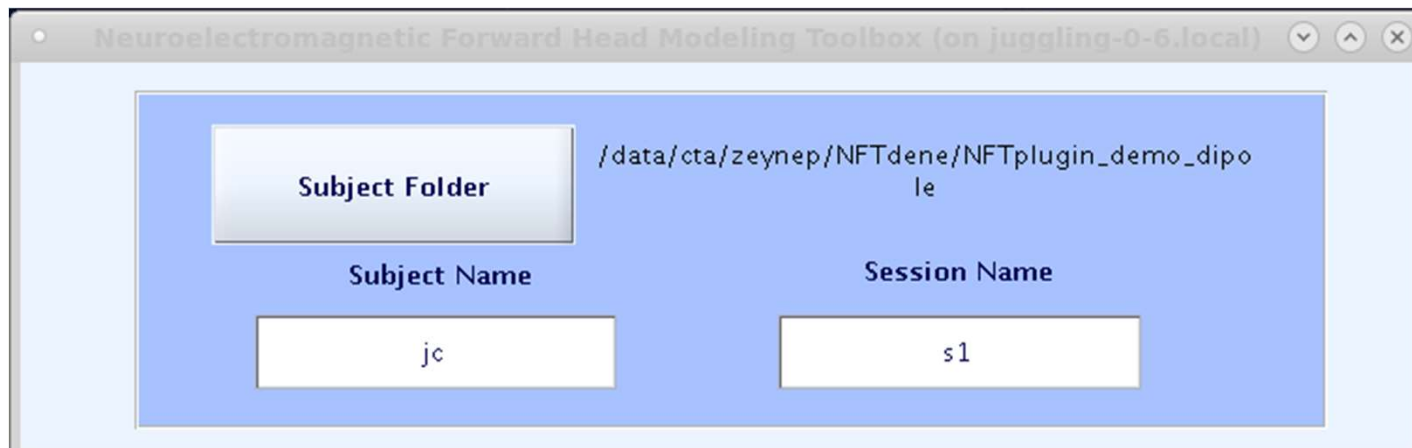
<https://rdi-share.ucsd.edu/message/U7F0uMivgbGZJur64TXDac>

Subject Selection

Subject Folder	
Subject Name	Session Name

- ◆ Select subject folder
- ◆ Specify subject name
- ◆ Specify session name

Subject Selection



The screenshot shows a software window titled "Neuroelectromagnetic Forward Head Modeling Toolbox (on juggling-0-6.local)". Inside the window, there is a blue panel with the following elements:

- A "Subject Folder" button with a light blue gradient.
- A text field containing the path: `/data/cta/zeynep/NFTdene/NFTplugin_demo_dipole`.
- A "Subject Name" label above a text input field containing the text "jc".
- A "Session Name" label above a text input field containing the text "s1".

Select current folder as subject folder
Enter "jc" as subject name
Enter "s1" as the session name

Image Segmentation

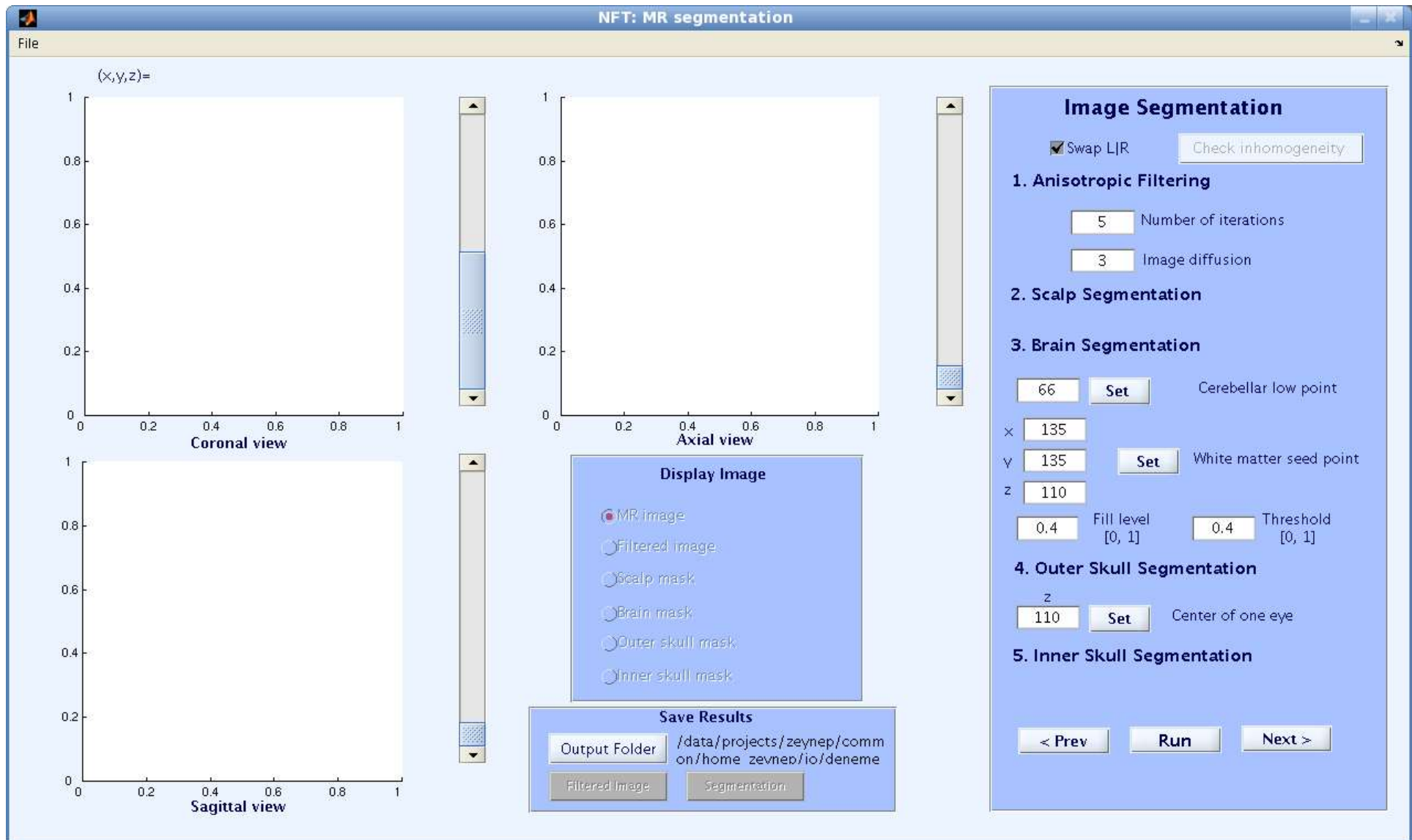
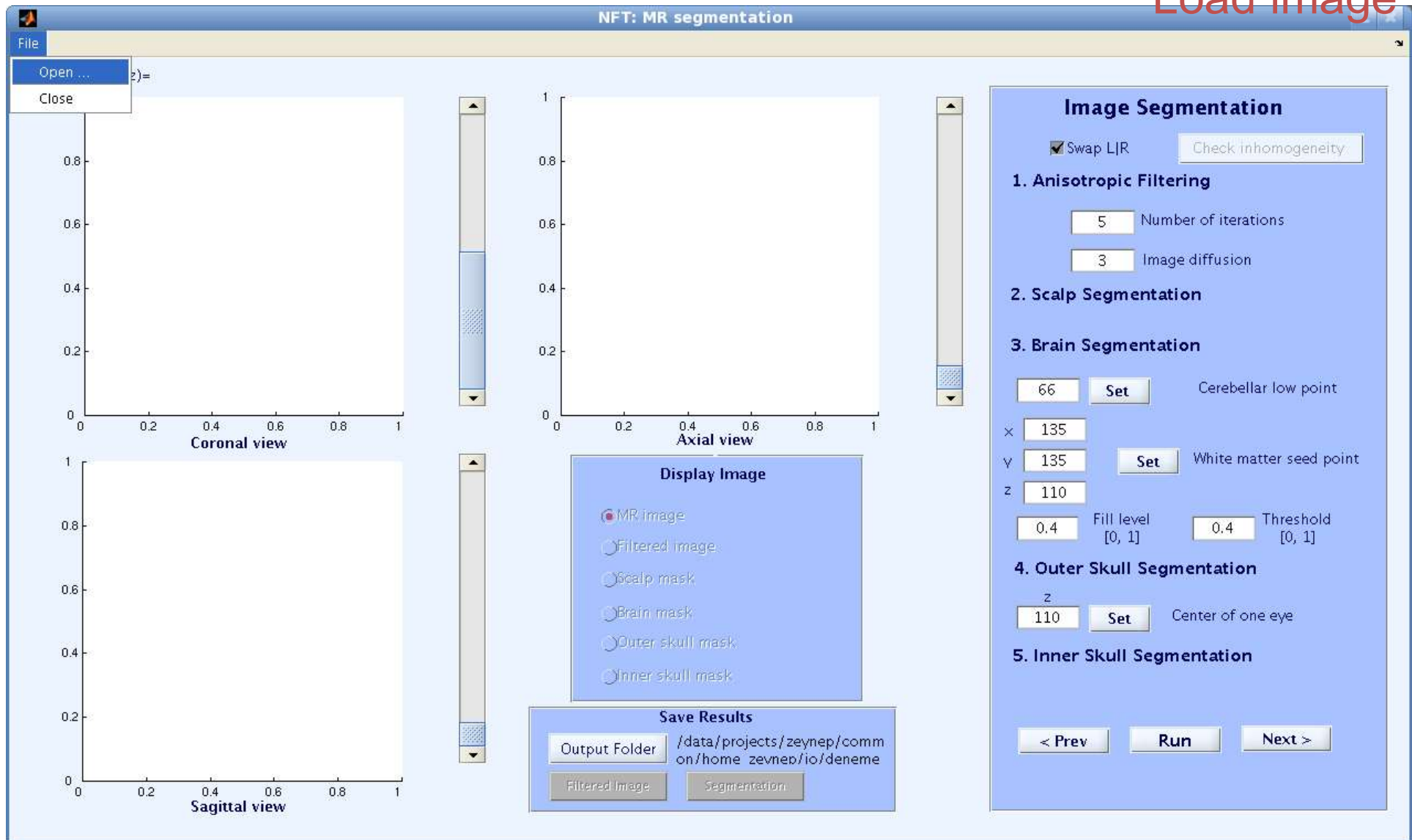


Image Segmentation

Load image



Segmentation

Load image test2c.img

NFT: MR segmentation

File

(x,y,z)=

Coronal view

Sagittal view

Select File to Open

Look In: jo

deneme, deneme_r, electrode, electrode2, electrode_reduced, fem, FS, jonton, mag, MNIdeneme, MNIdeneme2, MNlquad, NFT-ani, sil, skull_cond, symm, jo_ori.hdr, mri.hdr

File Name: mri.hdr

Files of Type: (*.hdr)

Open, Cancel

Open selected file

Outer skull mask, Inner skull mask

Save Results

Output Folder: /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image, Segmentation

Image Segmentation

☒ Swap L/R, Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations, 3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

66 Set Cerebellar low point

x 135, y 135 Set White matter seed point, z 110

0.4 Fill level [0, 1], 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

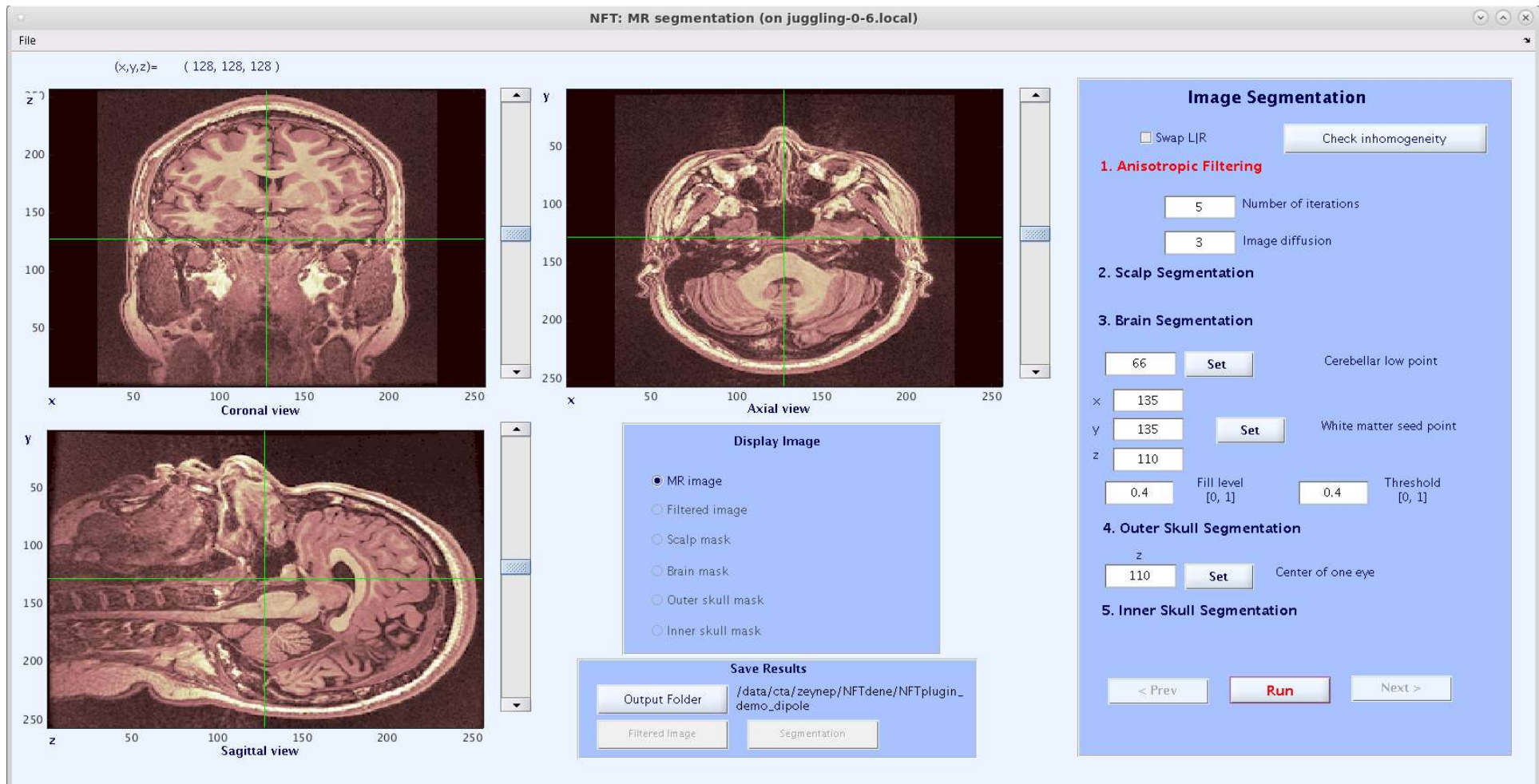
z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev, Run, Next >

Segmentation

Run filtering



Segmentation

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (128, 128, 128)

Coronal view

Axial view

Sagittal view

Display Image

- ☒ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

66 Set Cerebellar low point

x 135

y 135 Set White matter seed point

z 110

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

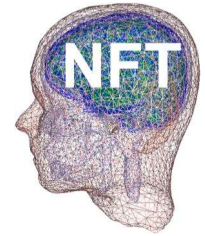
z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Run anisotropic filtering

Segmentation



View filtered image

NFT: MR segmentation (on juggling-0-6.local)

File (x,y,z)= (128, 128, 128)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image Segmentation

Image Segmentation

☐ Swap L/R Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

66 Set Cerebellar low point

x 135 y 135 z 110 Set White matter seed point

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Image is filtered!

Segmentation

Click 'Next' for scalp segmentation and run scalp segmentation

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (128, 128, 128)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

66 Set Cerebellar low point

x 135

y 135 Set White matter seed point

z 110

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

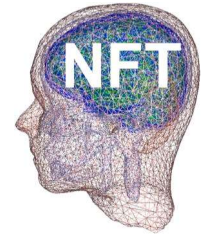
z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Image is filtered!

Segmentation



View scalp mask

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (128, 128, 128)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☒ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdne/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

66 Set Cerebellar low point

x 135 y 135 z 110 Set White matter seed point

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Scalp segmented!

Segmentation

Click 'Next' for brain segmentation
Selection of cerebellar low point

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (128, 176, 92)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdne/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Set Cerebellar low point

135 White matter seed point

135 Set

110

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Scalp segmented!

Segmentation

Selection of a white matter point

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (149, 176, 180)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Set Cerebellar low point

x 176

y 180 Set White matter seed point

z 149

0.4 Fill level [0, 1] 0.4 Threshold [0, 1]

4. Outer Skull Segmentation

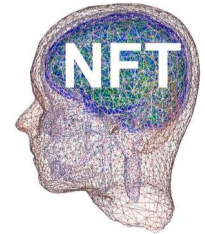
z 110 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Scalp segmented!

Segmentation



View brain mask

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (149, 176, 180)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☒ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image Segmentation

Image Segmentation

☐ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Cerebellar low point

x 176 White matter seed point

y 180

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 110 Center of one eye

5. Inner Skull Segmentation

< Prev **Run** Next >

Brain segmented!

Segmentation

Click 'Next' for skull segmentation

NFT: MR segmentation

File

(x,y,z) = (150, 172, 158)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☒ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Cerebellar low point

x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 110 Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Brain segmented!

Segmentation

Select a slice for eyes and click 'set'

NFT: MR segmentation (on juggling-0-6.local)

(x,y,z)= (149, 68, 142)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L|R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Set Cerebellar low point

x 176

y 180 Set White matter seed point

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 142 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Brain segmented!

Segmentation

Click 'Run' for skull segmentation

NFT: MR segmentation (on juggling-0-6.local)

(x,y,z)= (149, 68, 142)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☒ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Set Cerebellar low point

176 x

180 y Set White matter seed point

149 z

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

142 z Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Segmenting skull...

Segmentation

Click on the eyes

NFT: MR segmentation

Figure 2

(x,y,z)= (150, 67, 95)

Coronal view

Sagittal view

Brain mask
Outer skull mask
Inner skull mask

Save Results

Output Folder /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Set Cerebellar low point

172 x

158 y Set White matter seed point

150 z

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

95 z Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Segmenting skull...

Segmentation

NFT: MR segmentation

Figure 2

(x,y,z)= (150, 67, 95)

Coronal view

Sagittal view

Brain mask
Outer skull mask
Inner skull mask

Save Results

Output Folder /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Set Cerebellar low point

172

158 Set White matter seed point

150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

95 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Segmenting skull...

Segmentation

View skull segmentation

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (149, 68, 142)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☒ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder /data/cta/zeynep/NFTdne/NFTplugin_demo_dipole

Filtered Image Segmentation

Image Segmentation

☐ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Cerebellar low point

x 176

y 180 White matter seed point

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 142 Center of one eye

5. Inner Skull Segmentation

< Prev **Run** Next >

Skull segmented!

Segmentation

Click 'Next' for CSF segmentation

NFT: MR segmentation (on juggling-0-6.local)

(x,y,z)= (149, 68, 142)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☒ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image

Segmentation

Image Segmentation

☐ Swap L/R

Check inhomogeneity

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Set Cerebellar low point

x 176

y 180 Set White matter seed point

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 142 Set Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Skull segmented!

Segmentation

Click 'Run' for CSF segmentation

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (149, 68, 142)

z

200

150

100

50

x

50 100 150 200 250

Coronal view

y

50

100

150

200

250

x

50 100 150 200 250

Axial view

Y

50

100

150

200

250

z

50 100 150 200 250

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☒ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder /data/cta/zeypn/NFTdne/NFTplugin_demo_dipole

Filtered Image Segmentation

Image Segmentation

☐ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Cerebellar low point

x 176

y 180 White matter seed point

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 142 Center of one eye

5. Inner Skull Segmentation

< Prev **Run** Next >

Segmenting CSF...

Segmentation

NFT: MR segmentation (on juggling-0-6.local)

File

(x,y,z)= (149, 68, 142)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☒ Outer skull mask
- ☐ Inner skull mask

Save Results

Output Folder: /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Filtered Image Segmentation

Image Segmentation

☐ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

92 Cerebellar low point

x 176

y 180 White matter seed point

z 149

0.35 Fill level [0, 1] 0.35 Threshold [0, 1]

4. Outer Skull Segmentation

z 142 Center of one eye

5. Inner Skull Segmentation

< Prev **Run** Next >

Correcting skull and scalp...

Segmentation

View CSF segmentation

NFT: MR segmentation

(x,y,z) = (150, 160, 139)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☒ Inner skull mask

Save Results

Output Folder: /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Cerebellar low point

x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 95 Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Segmentation complete!

Segmentation

Save filtered image

NFT: MR segmentation

File

(x,y,z)= (150, 160, 139)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☒ Inner skull mask

Save Results

Output Folder /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Cerebellar low point

x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 95 Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Saving filtered image as SubjectA_filtered.mat

Segmentation

NFT: MR segmentation

File

(x,y,z)= (150, 160, 139)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☒ Inner skull mask

Save Results

Output Folder /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

☒ Swap L/R

1. Anisotropic Filtering

5 Number of iterations

3 Image diffusion

2. Scalp Segmentation

3. Brain Segmentation

67 Cerebellar low point

x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 95 Center of one eye

5. Inner Skull Segmentation

< Prev **Run** Next >

Filtered image saved as SubjectA_filtered.mat

Segmentation

Save segmentation

NFT: MR segmentation

File

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Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
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- ☐ Outer skull mask
- ☒ Inner skull mask

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Output Folder /data/projects/zeynep/comm on/home zeynep/io/deneme

Filtered Image Segmentation

Image Segmentation

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67 Cerebellar low point

x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 95 Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Saving segmentation as SubjectA_segments.mat

Segmentation

NFT: MR segmentation

File

(x,y,z)= (150, 160, 139)

Coronal view

Axial view

Sagittal view

Display Image

- ☐ MR image
- ☐ Filtered image
- ☐ Scalp mask
- ☐ Brain mask
- ☐ Outer skull mask
- ☒ Inner skull mask

Save Results

Output Folder /data/projects/zeynep/comm on/home zevnep/io/deneme

Filtered Image Segmentation

Image Segmentation

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5 Number of iterations

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x 172

y 158 White matter seed point

z 150

0.1 Fill level [0, 1] 0.1 Threshold [0, 1]

4. Outer Skull Segmentation

z 95 Center of one eye

5. Inner Skull Segmentation

< Prev Run Next >

Segmentation saved as SubjectA_segments.mat

Image Segmentation

Command Window

```
>> load jc_mri
>> mri

mri =

    struct with fields:

        dim: [256 256 256]
        xgrid: [1×256 double]
        ygrid: [1×256 double]
        zgrid: [1×256 double]
        anatomy: [256×256×256 double]
        transform: [4×4 double]
        hdr: 'test2c.hdr'

>> load jc_segments
>> Segm

Segm =

    struct with fields:

        scalpmask: [256×256×256 logical]
        brainmask: [256×256×256 logical]
        outerskullmask: [256×256×256 logical]
        innerskullmask: [256×256×256 logical]
        parameters: [1×1 struct]
```


Mesh Generation

From a magnetic
Resonance Image

Image Segmentation

Mesh Generation

Source Space Generation

Electrode Co-Registrati...

NFT: Mesh generation (on juggling-0-6.local)

Load Segmentation /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_segments

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

4 # of layers Mesh name: jc

☒ Linear ☐ Quadratic

7000 Number of nodes per layer

☐ Local mesh refinement

2.1 Edge length/
Distance between meshes

Start Mesh Generation

Status

Generate linear FEM mesh

Generate quadratic FEM mesh

Generate Mesh for a 3 or 4 layer head model

Mesh generation

Click local mesh refinement

NFT: Mesh generation (on juggling-0-6.local)

Load Segmentation /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_segments

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

of layers

Mesh name:

☒ Linear ☐ Quadratic

Number of nodes per layer

☒ Local mesh refinement

Edge length/
Distance between meshes

Start Mesh Generation

Status

Generate linear FEM mesh **Generate quadratic FEM mesh**

Mesh generation

NFT: Mesh generation (on juggling-0-6.local)

Load Segmentation /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_segments

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

of layers

Mesh name:

☒ Linear ☐ Quadratic

Number of nodes per layer

☒ Local mesh refinement

Edge length/
Distance between meshes

Start Mesh Generation

Mesh saved!

Generate linear FEM mesh **Generate quadratic FEM mesh**

Mesh generation

NFT: Mesh generation (on juggling-0-6.local)

Load Segmentation /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_segments

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

of layers

Mesh name:

☒ Linear ☐ Quadratic

Number of nodes per layer

☒ Local mesh refinement

Edge length/
Distance between meshes

Start Mesh Generation

Mesh saved!

Generate linear FEM mesh **Generate quadratic FEM mesh**

- ◆ `[C,E] = ReadSMF('Scalp.smf',0,0,0,1);`
- ◆ `Plotmesh(E(:,2:4),C(:,2:4))`

Mesh generation

NFT: Mesh generation (on juggling-0-6.local)

Load Segmentation /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_segments

Output Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

of layers

Mesh name:

☒ Linear ☐ Quadratic

Number of nodes per layer

☒ Local mesh refinement

Edge length/
Distance between meshes

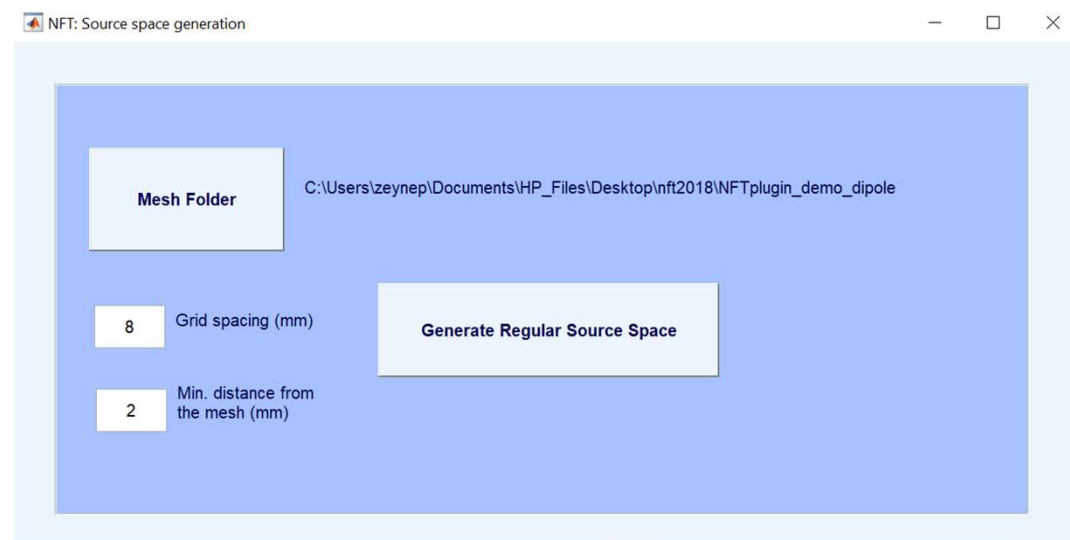
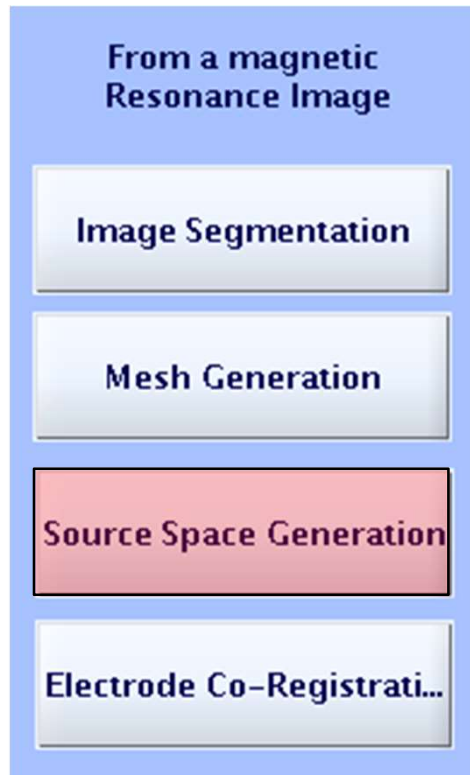
Start Mesh Generation

Linear FEM mesh generated!

Generate linear FEM mesh

Generate quadratic FEM mesh

Source Space Generation



Generates a simple source space:
Regular Grid inside the brain
With a given spacing and distance to the mesh

Source Space Generation

From a magnetic
Resonance Image

Image Segmentation

Mesh Generation

Source Space Generation

Electrode Co-Registrati...

NFT: Source space generation

Mesh Folder C:\Users\zeynep\Documents\HP_Files\Desktop\nft2018\NFTplugin_demo_dipole

8 Grid spacing (mm)

2 Min. distance from the mesh (mm)

Generate Regular Source Space

Source space saved!

Electrode Co-registration

From a magnetic
Resonance Image

Image Segmentation

Mesh Generation

Source Space Generation

Electrode Co-Registrati...

NFT: Electrode co-registration (on juggling-0-6.local)

Load sensor locations

Mesh Folder

Initial co-registration

Complete co-registration

Save initial reg.

Electrode Co-registration

From a magnetic Resonance Image

Image Segmentation

Mesh Generation

Source Space Generation

Electrode Co-Registrati...

NFT: Electrode co-registration (on juggling-0-6.local)

Load sensor locations Electrode file name

Mesh Folder /data/cta/zeynep/NFTdene/NFTplugin_demo_dipole

Initial co-registration

Complete co-registration

Select File to Open (on juggling-0-6.local)

Look In: NFTplugin_demo_dipole

- jc.1.node
- jc.bec
- jc.bee
- jc.bei
- jc.smesh
- jc_fid.sfp**
- jc_mesh.mat
- jc_mri.mat

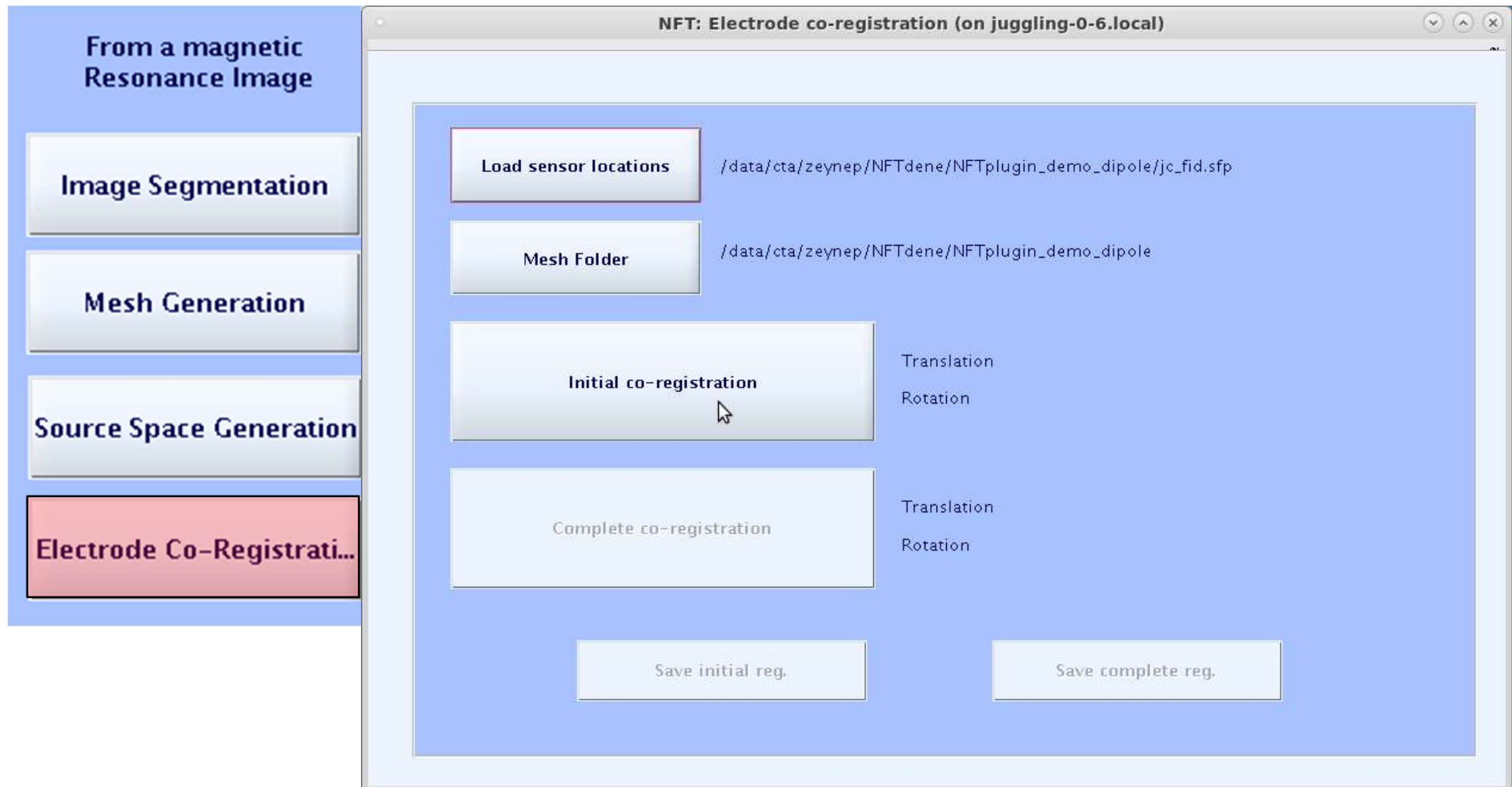
File Name: jc_fid.sfp

Files of Type: All Files

Open Cancel

Save initial reg.

Electrode Co-registration



Electrode Co-registration

From a magnetic Resonance Image

Image Segmentation

Mesh Generation

Source Space Generation

Electrode Co-Registration

NFT: Electrode co-registration (on juggling-0-6.local)

coregister() (on juggling-0-6.local)

File Edit View Insert Tools Desktop Window Help

Labels on
Electrodes
Labels on
Electrodes
Mesh off

Help me
Funct. h...

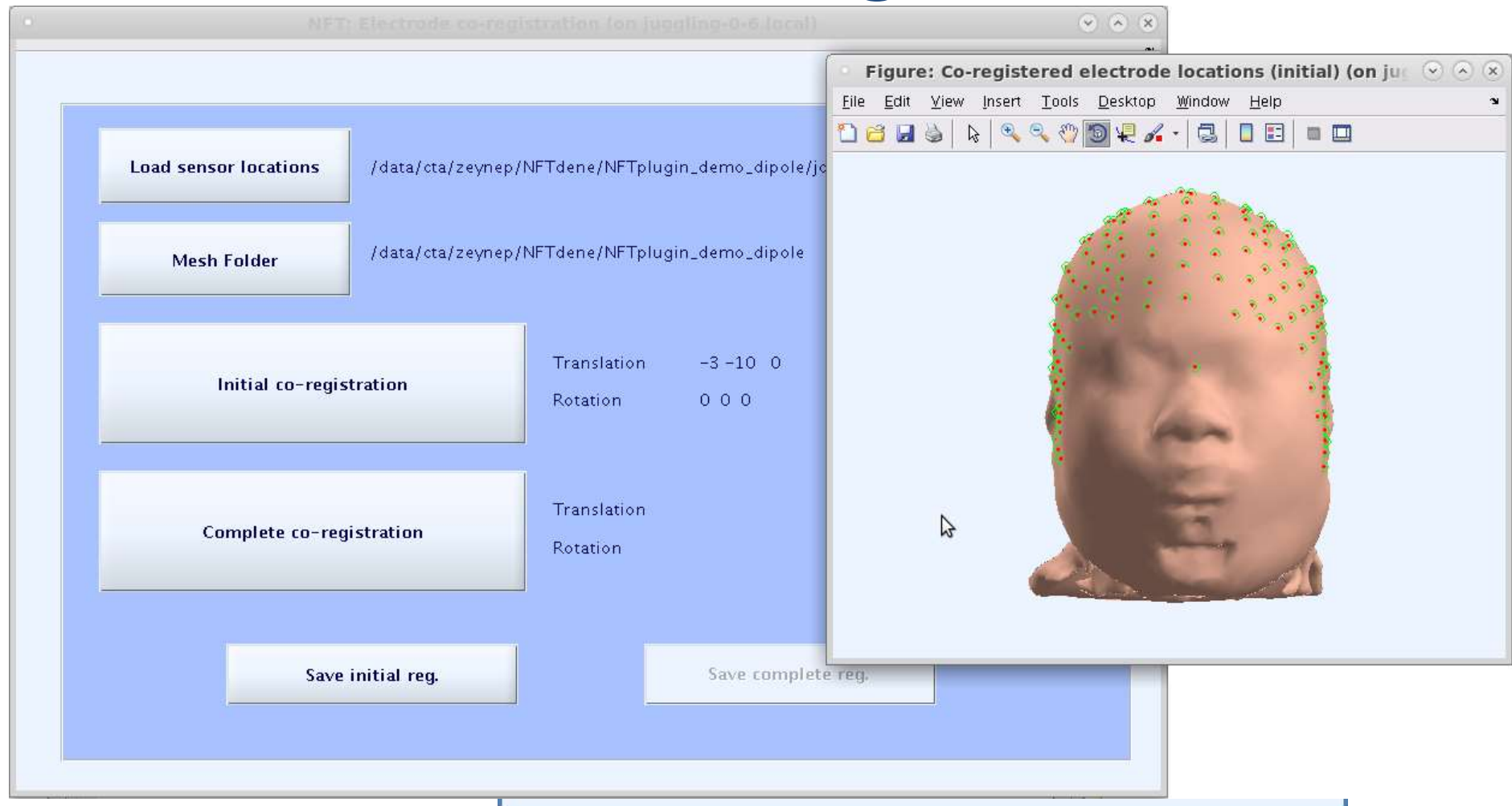
Move right 0 Pitch (rad) 0 Resize (x) 1 Align fiducials
Move front 0 Roll (rad) 0 Resize (y) 1 Warp montage
Move up (mm) 0 Yaw (rad) 0 Resize (z) 1 Cancel Ok

Save initial reg. Save complete reg.

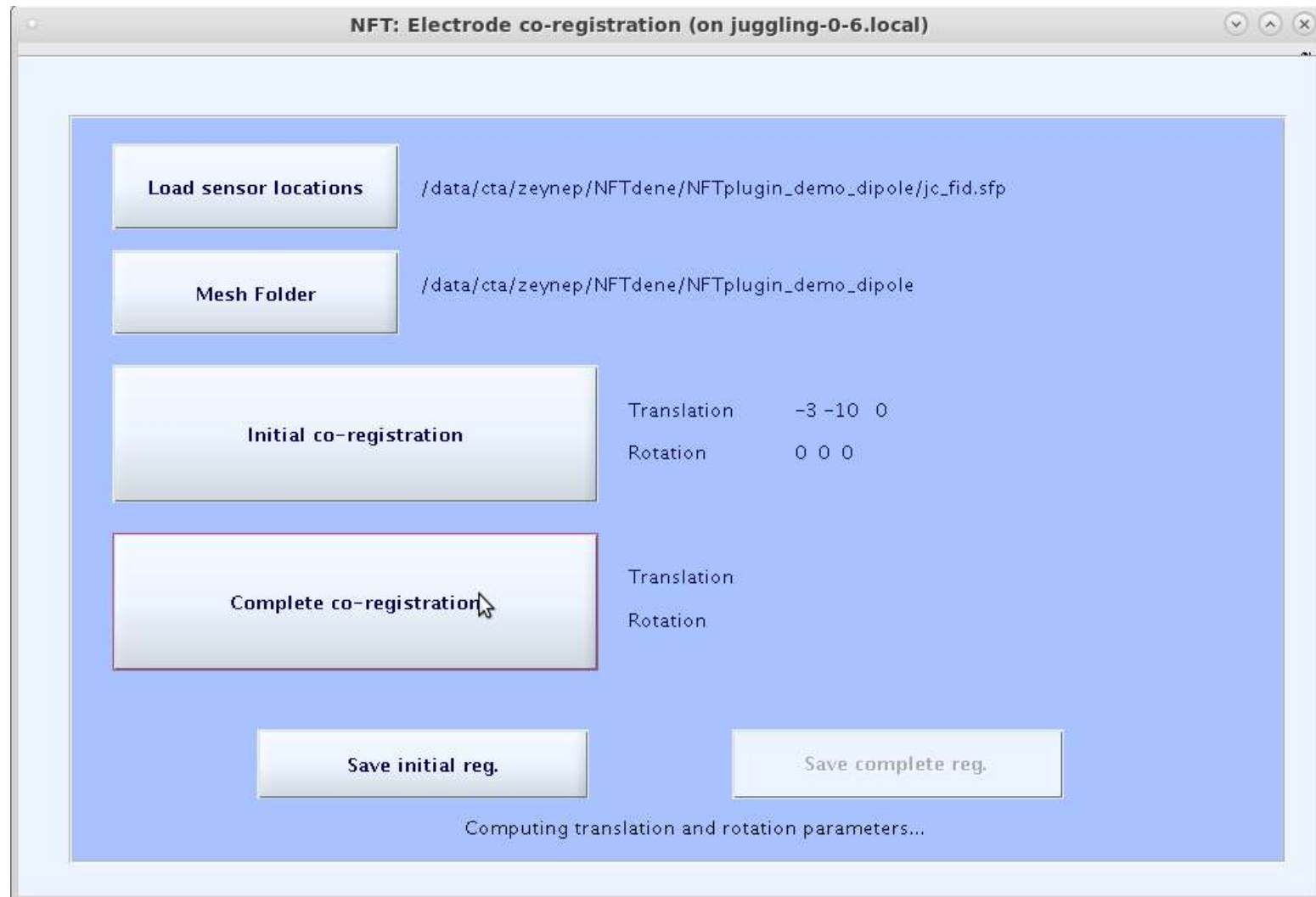
Load sensor locations /data/
Mesh Folder /data/
Initial co-registration
Complete co-registration

Save initial reg. Save complete reg.

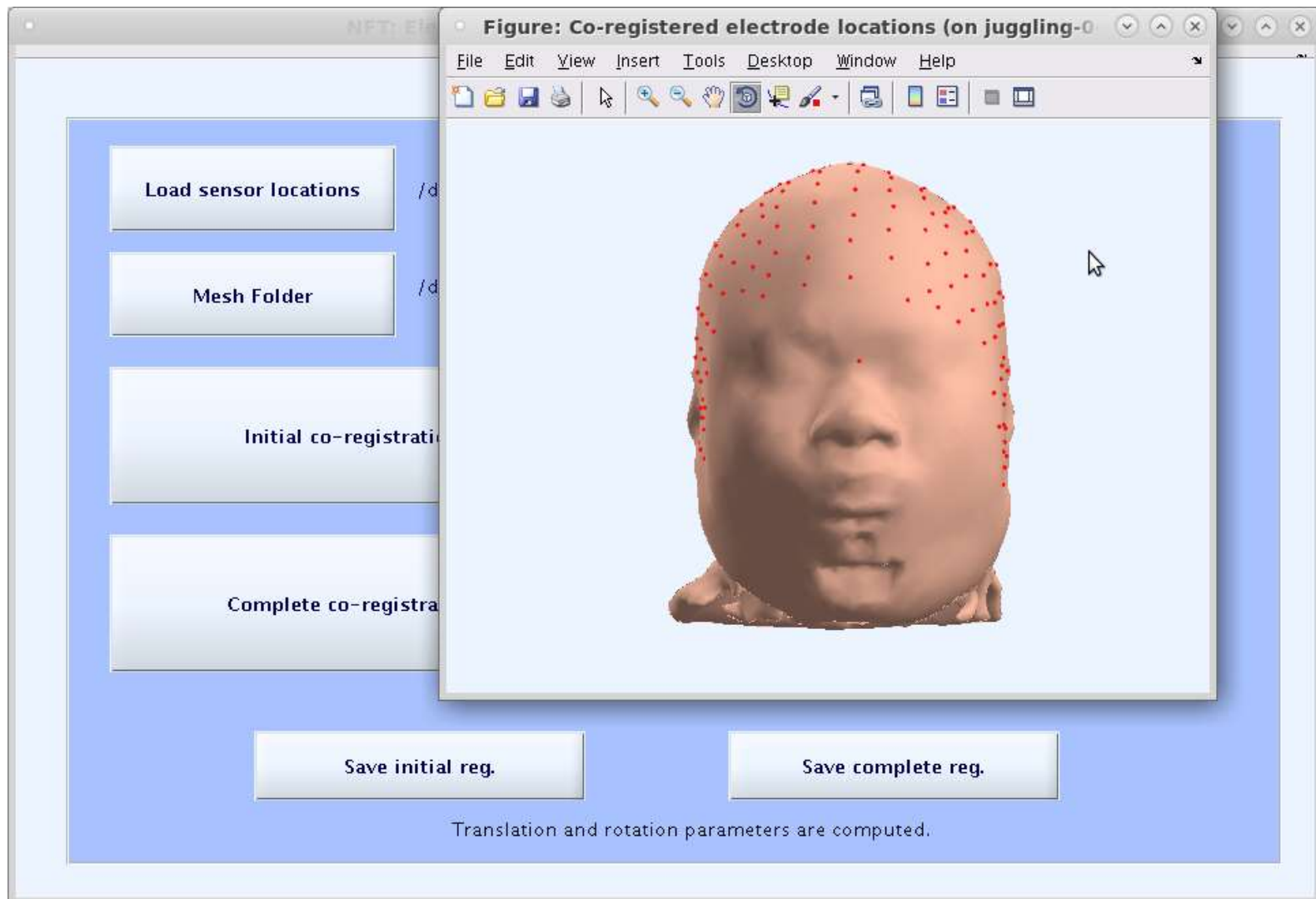
Electrode Co-registration



Electrode co-registration



Electrode co-registration

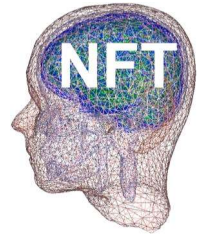


Mesh generation

NFT: Electrode co-registration (on juggling-0-6.local)

Load sensor locations	/data/cta/zeynep/NFTdene/NFTplugin_demo_dipole/jc_fid.sfp
Mesh Folder	/data/cta/zeynep/NFTdene/NFTplugin_demo_dipole
Initial co-registration	Translation -3 -10 0 Rotation 0 0 0
Complete co-registration	Translation 10.3303 -1.18985 -11.5649 Rotation 1.6886 -3.3001 0.42156
Save initial reg.	Save complete reg.

Automatic registration is saved.



```
>> sens=load('jc_s1.sensors','-mat')
```

```
sens =
```

```
    fn: '/data/cta/zeynep/NFTdene/JC/jc_fid.sfp'  
    eloc: [1x208 struct]  
    pnt: [208x3 double]  
    ind: [1x208 double]  
    param: [1x1 struct]
```

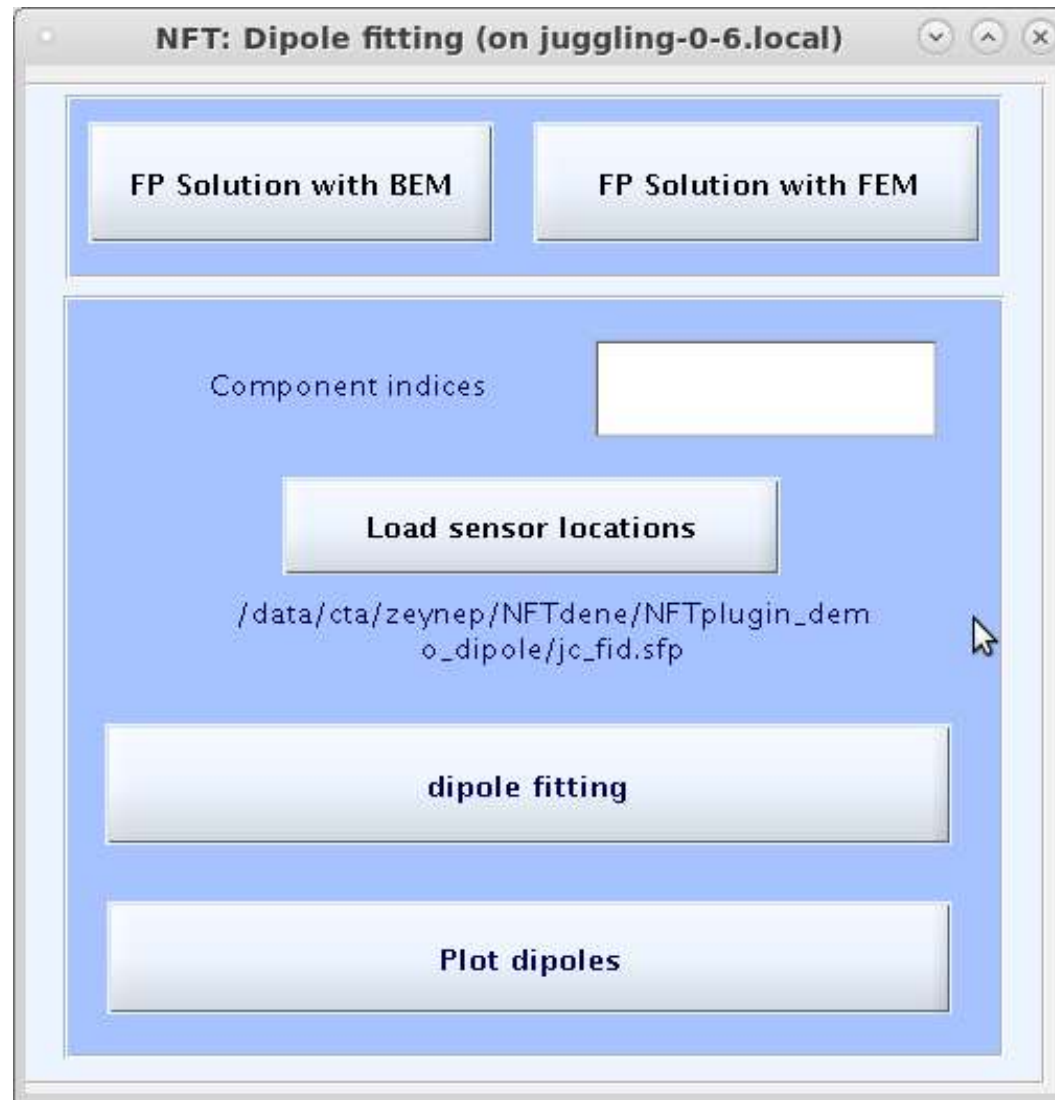
```
>> sens.param
```

```
ans =
```

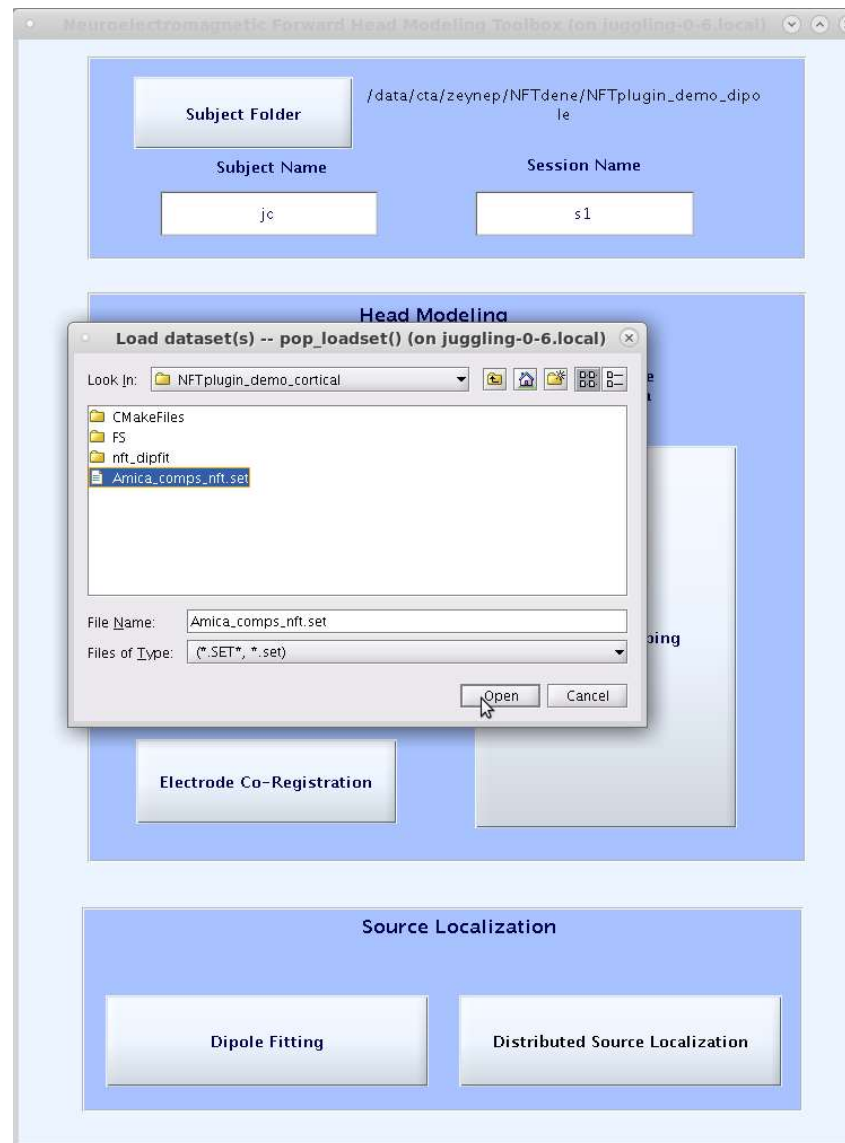
```
    init: [-3 -10 0 0 0 0 1 1 1]  
    auto: [10.3303 -1.1899 -11.5649 1.6886 -3.3001 0.4216]
```

```
>> |
```

Dipole source localization



Select EEG data



Forward Problem Solver

- ◆ MATLAB interface to numerical solvers
- ◆ Boundary Element Method or Finite Element Method
 - EEG Only (for now)
 - Interfaces to the Matrix generator executable written in C++
- ◆ Other computation done in MATLAB
- ◆ Generated matrices are stored on disk for future use.

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

Mesh Name

Number of Layers

Number of Nodes

Number of Elements

Number of Nodes/Element

BEM Model

Model Name

Enter conductivity values:

Scalp Skull

Brain CSF

☒ Modified (Isolated Problem Approach)

Value Changed!

Session

Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Value Changed!

Forward Problem Solution

For Dipole

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

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Number of Elements

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Scalp Skull

Brain CSF

☒ Modified (Isolated Problem Approach)

Value Changed!

Session

Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Value Changed!

Forward Problem Solution

For Dipole

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

jc Mesh Name

Show Mesh

4 Number of Layers

16151 Number of Nodes

32286 Number of Elements

3 Number of Nodes/Element

BEM Model

jc Model Name

Enter conductivity values:

0.33 Scalp 0.0132 Skull

0.33 Brain 1.79 CSF

☒ Modified (Isolated Problem Approach)

Create Model

BEM Model Created

Session

s1 Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Load

Show Sensors

Generate transfer matrix

Value Changed!

Forward Problem Solution

Load Source Space

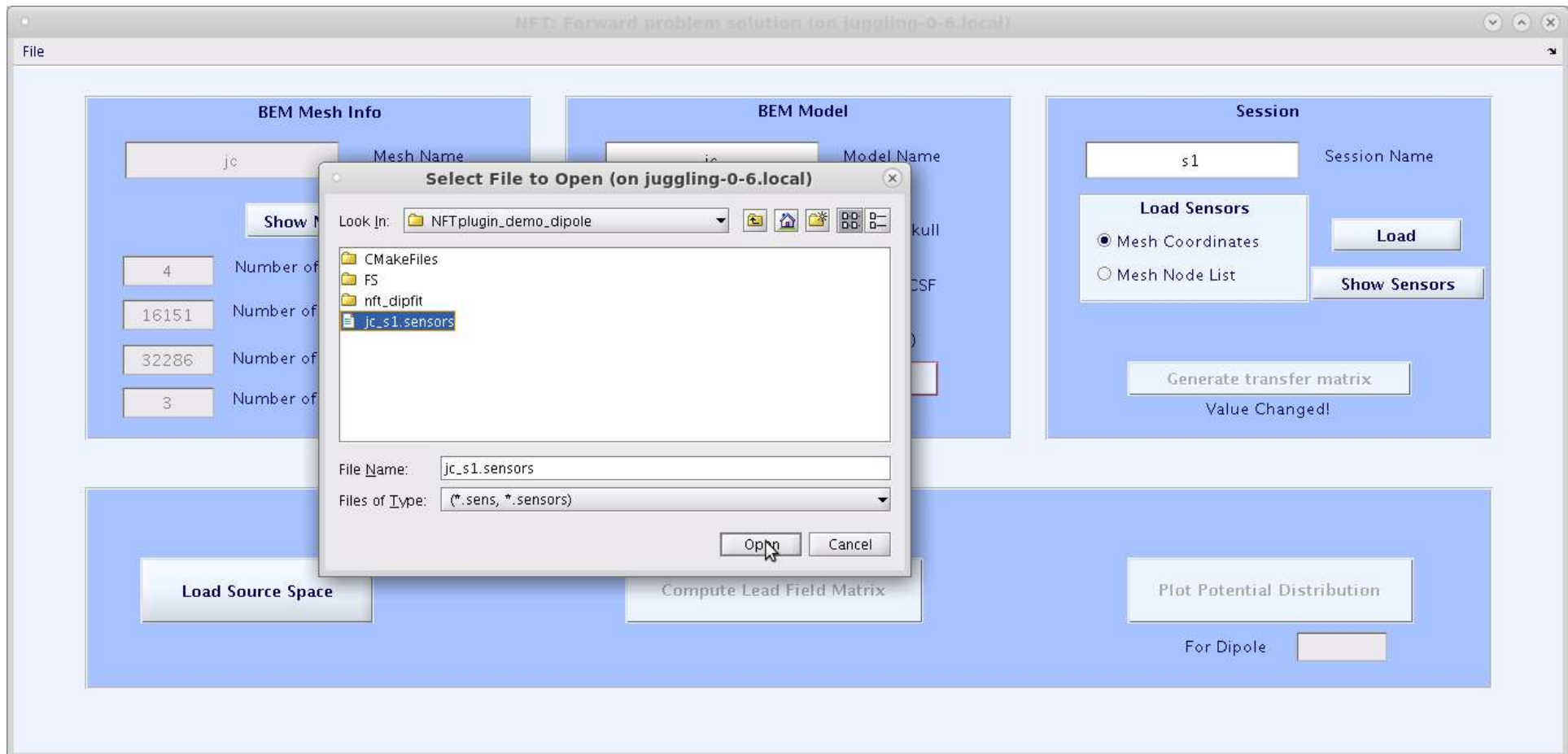
Compute Lead Field Matrix

Plot Potential Distribution

For Dipole

Forward Problem Solution with BEM

Forward Model Generation



Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

Mesh Name:

Show Mesh

Number of Layers:

Number of Nodes:

Number of Elements:

Number of Nodes/Element:

BEM Model

Model Name:

Enter conductivity values:

Scalp: Skull:

Brain: CSF:

☒ Modified (Isolated Problem Approach)

Create Model

BEM Model Loaded

Session

Session Name:

Load Sensors

☒ Mesh Coordinates ☐ Mesh Node List

Load

Show Sensors

Sensors Loaded:

Generate transfer matrix

Session Loaded

Forward Problem Solution

Load Source Space

Compute Lead Field Matrix

Plot Potential Distribution

For Dipole:

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

Mesh Name:

Show Mesh

Number of Layers:

Number of Nodes:

Number of Elements:

Number of Nodes/Element:

Enter conductivity:

Enter resistivity:

☒ Modified (Isotropic)

Session Name:

Load

Show Sensors

Sensors Loaded:

matrix

ed:

Select File to Open (on juggling-0-6.local)

Look In:

- CMakeFiles
- FS
- nft_dipfit
- jc_sourcepace.dip**

File Name:

Files of Type:

Open **Cancel**

Forward Problem Solution

Load Source Space

Compute Lead Field Matrix

Plot Potential Distribution

For Dipole:

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

Mesh Name

Number of Layers

Number of Nodes

Number of Elements

Number of Nodes/Element

BEM Model

Model Name

Enter conductivity values:

Scalp Skull

Brain CSF

☒ Modified (Isolated Problem Approach)

BEM Model Loaded

Session

Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Sensors Loaded

Session Loaded

Forward Problem Solution

Dipoles Loaded

For Dipole

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

Mesh Name

Number of Layers

Number of Nodes

Number of Elements

Number of Nodes/Element

BEM Model

Model Name

Enter conductivity values:

Scalp Skull

Brain CSF

☒ Modified (Isolated Problem Approach)

BEM Model Loaded

Session

Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Sensors Loaded

Session Loaded

Forward Problem Solution

Dipoles Loaded

LFM Computed

For Dipole

Forward Problem Solution with BEM

Forward Model Generation

NFT: Forward problem solution (on juggling-0-6.local)

File

BEM Mesh Info

jc Mesh Name

Show Mesh

4 Number of Layers

16151 Number of Nodes

32286 Number of Elements

3 Number of Nodes/Element

BEM Model

jc Model Name

Enter conductivity values:

0.33 Scalp 0.0132 Skull

0.33 Brain 1.79 CSF

☒ Modified (Isolated Problem Approach)

Create Model

BEM Model Loaded

Session

s1 Session Name

Load Sensors

☒ Mesh Coordinates

☐ Mesh Node List

Load

Show Sensors

208 Sensors Loaded

Generate transfer matrix

Session Loaded

Forward Problem Solution

Load Source Space

7479 Dipoles Loaded

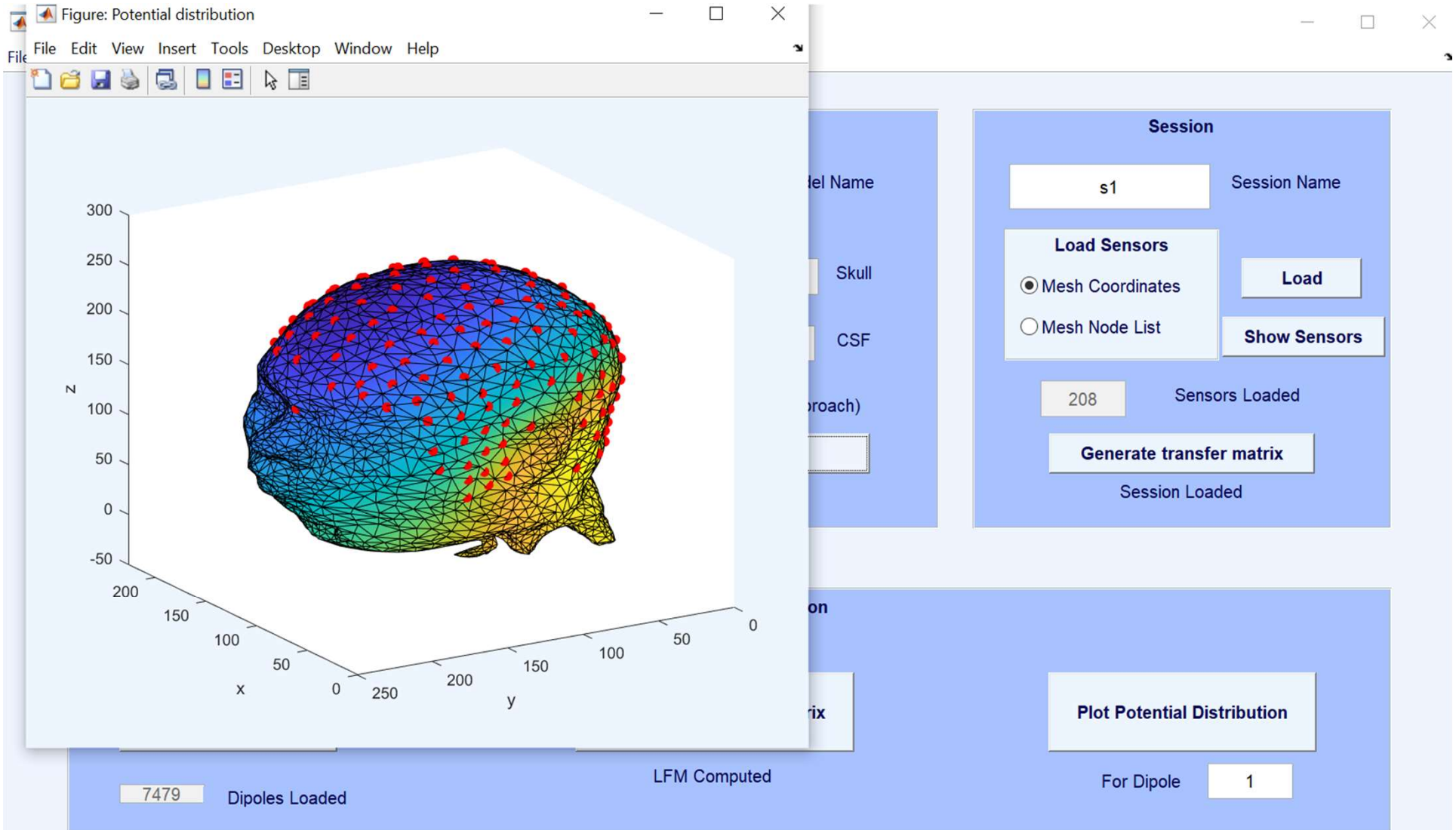
Compute Lead Field Matrix

LFM Computed

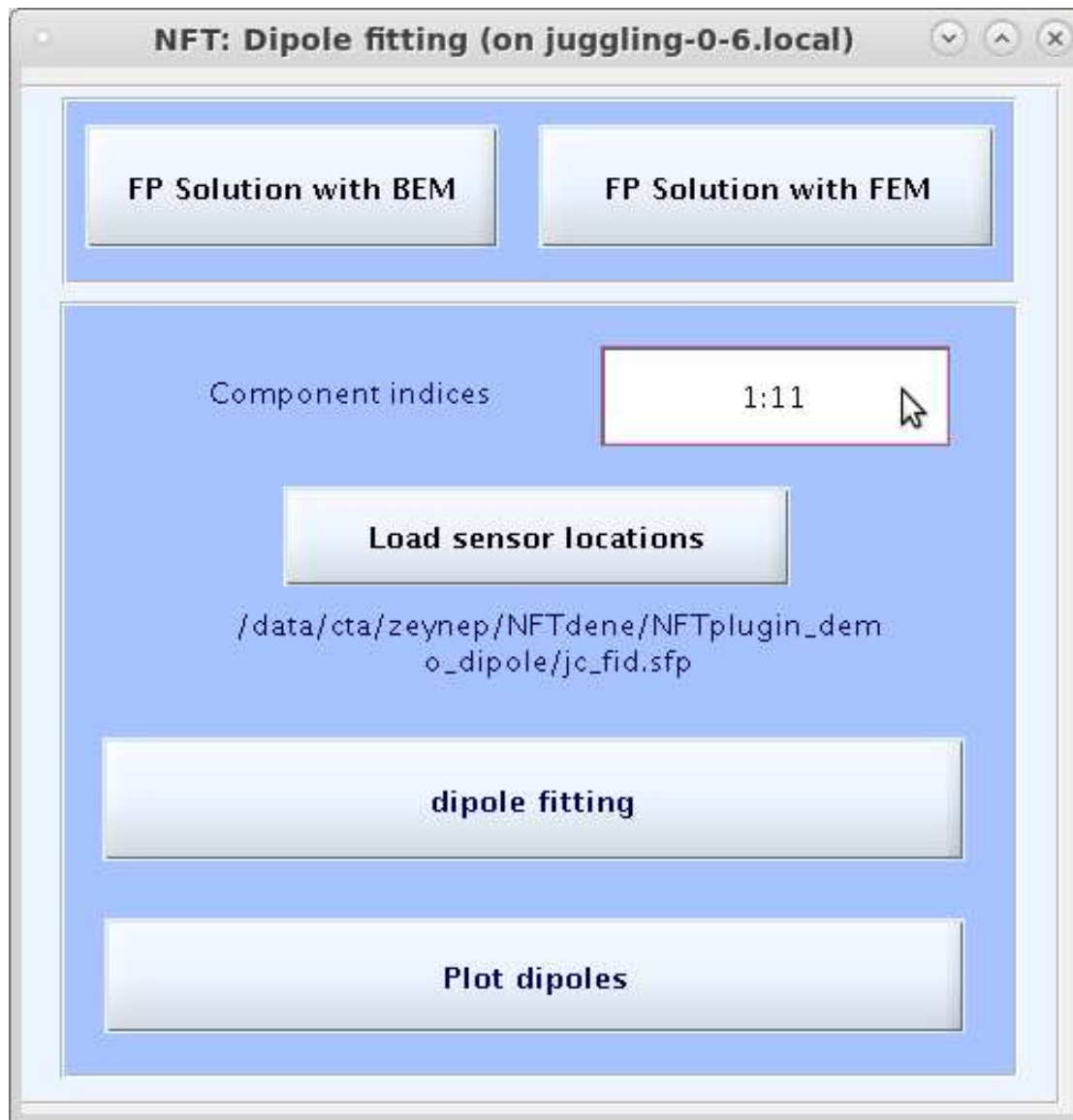
Plot Potential Distribution

For Dipole

Forward Problem Solution with BEM



Inverse Problem Solution with BEM



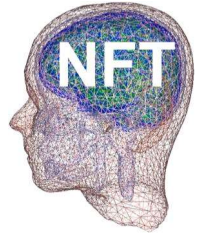
Output

- ◆ Dipole source localization is saved in EEG structure, under EEG.etc.nft.
- ◆ After source localization with NFT, you can continue using EEGLAB;

```
EEG.dipfit.model = EEG.etc.nft.model;
```

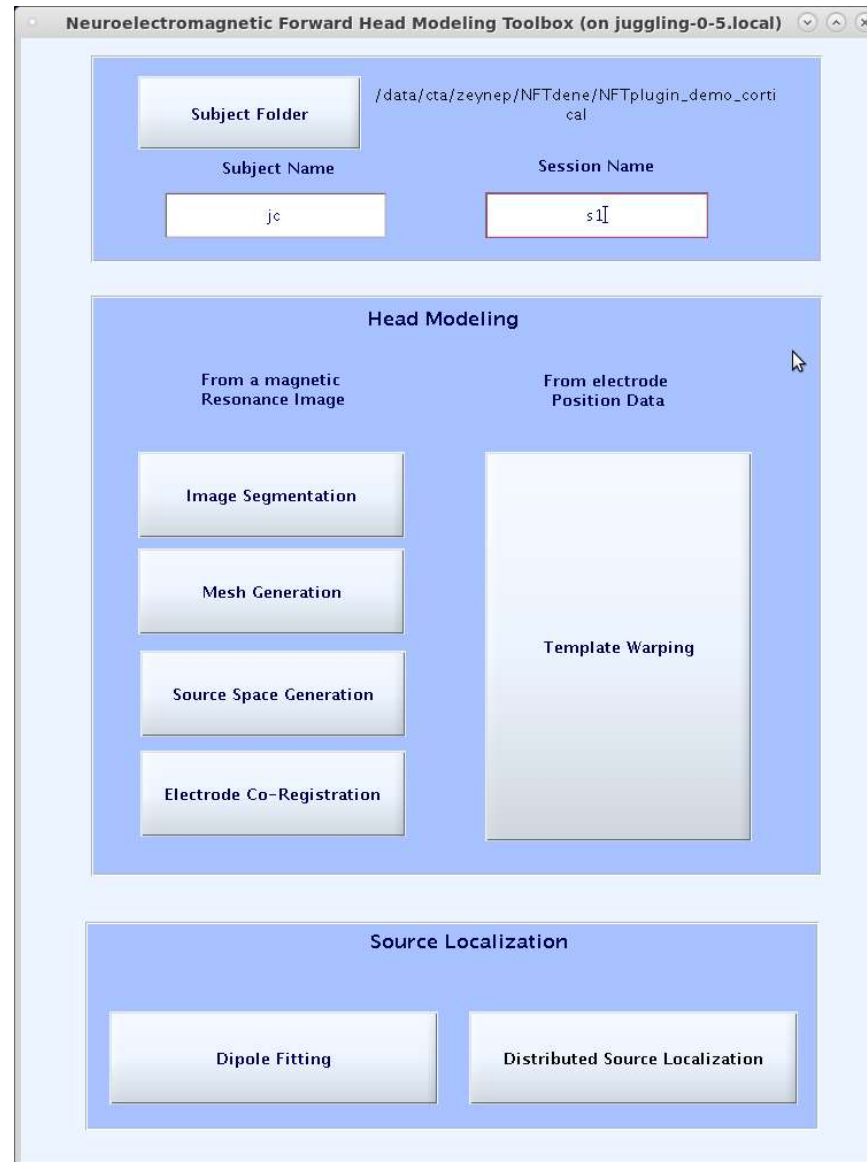


Distributed Source localization

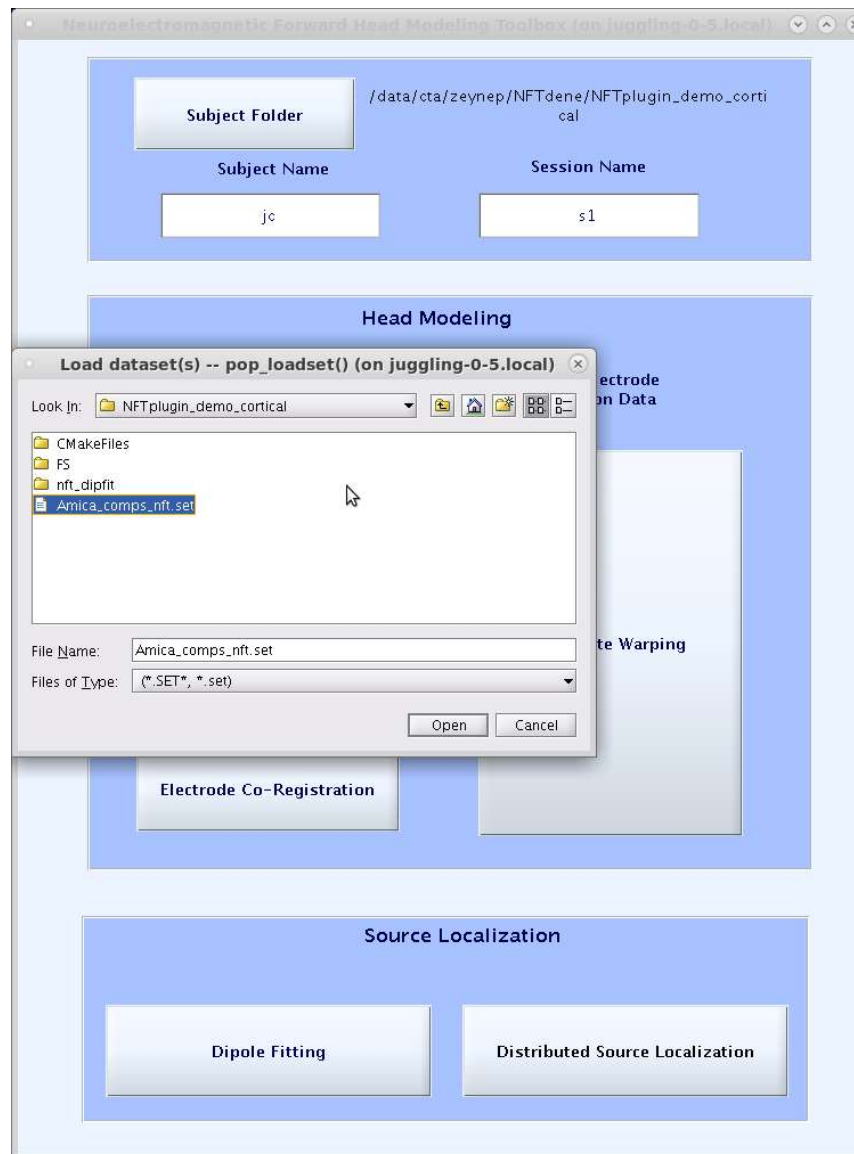


Go to the folder NFTplugin_demo_cortical

Distributed Source Localization

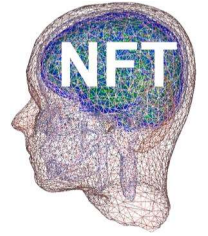


Select EEG data





NIST – Generation of a cortical source space



- ◆ Load Freesurface cortical surface
- ◆ Downsample to 80,000 vertices
- ◆ Co-register with the NFT brain surface
- ◆ Re-generate NFT head model
- ◆ Calculate normals for each vertex on the cortical surface
- ◆ Save the cortical source space as: **Subject_name FS_ss.dip**
- ◆ Calculate node area of each vertex for source localization, save as **Node_area**
- ◆ Check if the sensor locations need to be updated according to the new NFT model.

Distributed Source Localization

Distributed_Source_Localization (on juggling-0-5.local)

Load MRI MRI file

Start Freesurfer

Cortical source space 80000 10 mm Generate patches

of dipoles in source space

Forward Problem Solution

FP Solution with BEM FP Solution with FEM

Cortical Source Localization

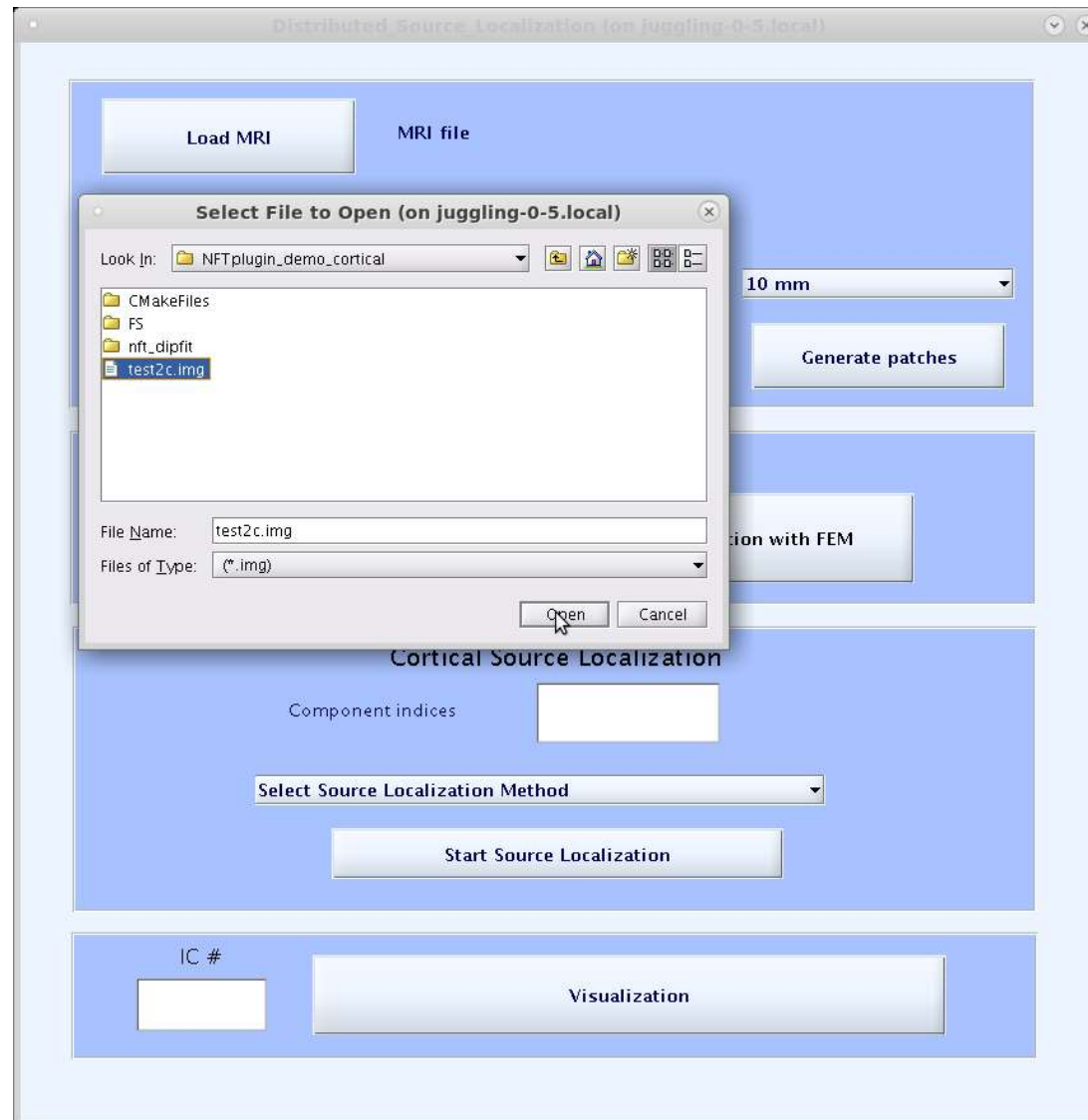
Component indices

Select Source Localization Method

Start Source Localization

IC # Visualization

Load MRI



Run Freesurfer

Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer

Cortical source space # of dipoles in source space

Generate patches

Forward Problem Solution

FP Solution with BEM

Cortical Source Localization

Component indices

Select Source Localization Method

Start Source Localization

IC #

Visualization

Freesurfer completed

Distributed Source Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer Running Freesurfer completed!

Cortical source space # of dipoles in source space 10 mm Generate patches

Forward Problem Solution

FP Solution with BEM FP Solution with FEM

Cortical Source Localization

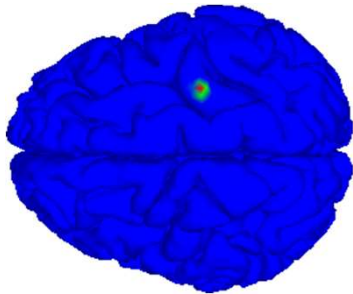
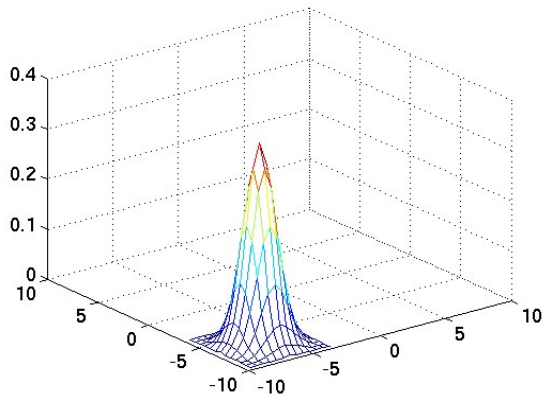
Component indices

Select Source Localization Method

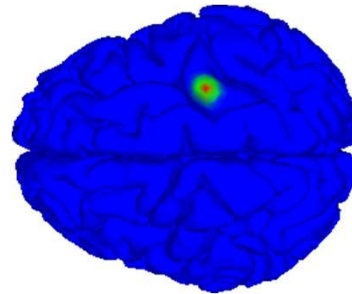
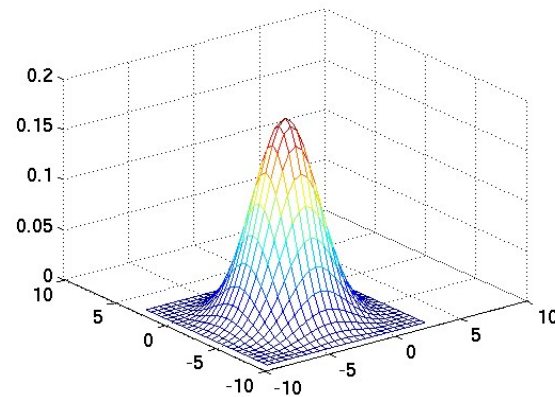
Start Source Localization

IC # Visualization

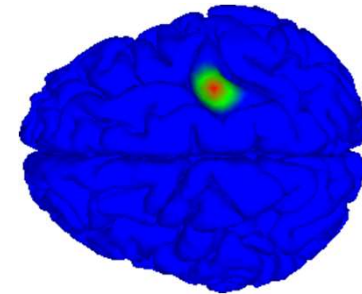
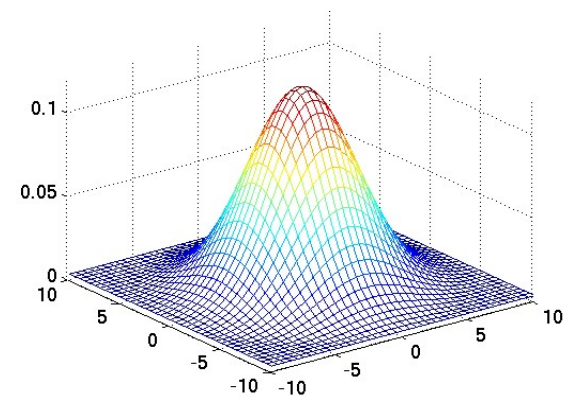
NIST – Patch generation



3 mm patch

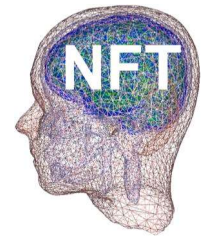


6 mm patch



10 mm patch

NIST has options to generate Gaussian patches with 10 mm, 6 mm, 3 mm in radius.



Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer Running Freesurfer completed!

Cortical source space # of dipoles in source space

3,6,10 mm

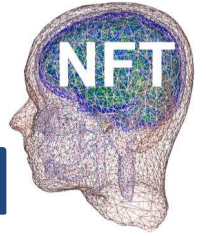
Forward Problem Solution

Cortical Source Localization

Component indices

Select Source Localization Method

IC #



Forward Problem Solution with FEM

- ◆ Tetgen for mesh generation
 - Uses BEM meshes as boundaries
- ◆ METU-FEM to generate transfer matrix
 - Compiled from source
 - Requires PETSc for matrix operations
- ◆ metufem .mex file for forward solutions in MATLAB
- ◆ Instructions available under README.FEM file.



NFT: Forward problem solution (on juggling-0-5.local)

File

Select File to Open (on juggling-0-5.local)

Look In: NFTplugin_demo_cortical

- CMaKeFiles
- FS
- nft_dipfit
- jc.1.msh
- jcFS.1.msh**

File Name: jcFS.1.msh

Files of Type: (*.msh)

Open

Cancel

FEM Session

jcFS

Session Name

for conductivity values:

Scalp

0.0132

Skull

Brain

Sensors

Sensors Loaded

Create Session

Value Changed!

Load Source Space

Compute Lead Field Matrix



NFT: Forward problem solution (on juggling-0-5.local)

File

FEM Mesh Info

jcFS.1.msh

Mesh Name

Show Mesh

FEM Session

Session Name

Enter conductivity values:

0.33

Scalp

0.0132

Skull

0.33

Brain

1.79

CSF

Load sensors

Sensors Loaded

Create Session

No Session

tion

Compute Lead Field Matrix

Select File to Open (on juggling-0-5.local)

Look In: NFTplugin_demo_cortical

- CMakeFiles
- FS
- nft_dipfit
- jc_s1.sensors

File Name: jc_s1.sensors

Files of Type: (*.sens, *.sensors)

Open

Cancel



NFT: Forward problem solution (on juggling-0-5.local)

File

FEM Mesh Info

jcFS.1.msh

Mesh Name

Show Mesh

4

Number of Layers

297956

Number of Nodes

4

Number of Nodes/Element

FEM Session

s1

Session Name

Enter conductivity values:

0.33

Scalp

0.0132

Skull

0.33

Brain

1.79

CSF

Load sensors

208

Sensors Loaded

Create Session

Value Changed!

Forward Problem Solution

Load Source Space

Compute Lead Field Matrix



NFT: Forward problem solution (on juggling-0-5.local)

File

Select File to Open (on juggling-0-5.local)

Look In: NFTplugin_demo_cortical

- CMaFiles
- FS
- nft_dipfit
- jc_sourcespace.dip
- jcFS_ss.dip

File Name: jcFS_ss.dip

Files of Type: (*.dip)

Open

Cancel

FEM Session

s1

Session Name

conductivity values:

Scalp

0.0132

Skull

Brain

1.79

CSF

Sensors

208

Sensors Loaded

Create Session

FEM Session Created

Forward Problem Solution

Load Source Space

Compute Lead Field Matrix



NFT: Forward problem solution (on juggling-0-5.local)

File

FEM Mesh Info

jcFS.1.msh

Mesh Name

Show Mesh

4

Number of Layers

297956

Number of Nodes

4

Number of Nodes/Element

FEM Session

s1

Session Name

Enter conductivity values:

0.33

Scalp

0.0132

Skull

0.33

Brain

1.79

CSF

Load sensors

208

Sensors Loaded

Create Session

FEM Session Created

Forward Problem Solution

Load Source Space

80150

Dipoles Loaded

Compute Lead Field Matrix



NFT: Forward problem solution (on juggling-0-5.local)

File

FEM Mesh Info

jcFS.1.msh

Mesh Name

Show Mesh

4

Number of Layers

297956

Number of Nodes

4

Number of Nodes/Element

FEM Session

s1

Session Name

Enter conductivity values:

0.33

Scalp

0.0132

Skull

0.33

Brain

1.79

CSF

Load sensors

208

Sensors Loaded

Create Session

FEM Session Loaded

Forward Problem Solution

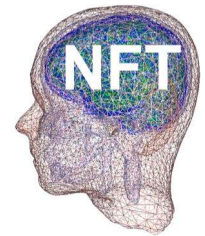
Load Source Space

80150

Dipoles Loaded

Compute Lead Field Matrix

LFM Computed



Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer

Cortical source space **Generate patches**
of dipoles in source space

Forward Problem Solution

FP Solution with BEM **FP Solution with FEM**

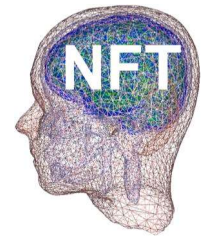
Cortical Source Localization

Component indices

Select Source Localization Method

Start Source Localization

IC # **Visualization**



Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer

Cortical source space **Generate patches**
of dipoles in source space

Forward Problem Solution

FP Solution with BEM **FP Solution with FEM**

Cortical Source Localization

Component indices

Sparse compact and smooth (SCS) method

Start Source Localization

IC # **Visualization**

Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer

Cortical source space **Generate patches**
of dipoles in source space

Forward Problem Solution

FP Solution with BEM **FP Solution with FEM**

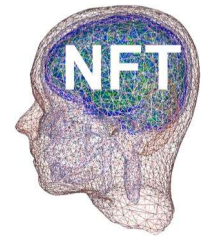
Cortical Source Localization

Component indices

Sparse compact and smooth (SCS) method

Start Source Localization

IC # **Visualization**



Distributed_Source_Localization (on juggling-0-5.local)

Load MRI

Start Freesurfer

Cortical source space **Generate patches**
of dipoles in source space

Forward Problem Solution

FP Solution with BEM **FP Solution with FEM**

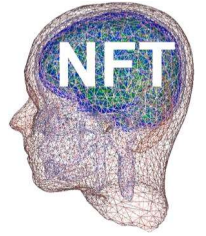
Cortical Source Localization

Component indices

Sparse compact and smooth (SCS) method

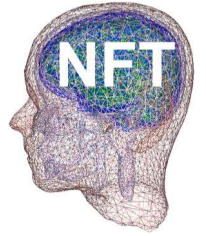
Start Source Localization

IC # **Visualization**



Output

- ◆ Source estimates are saved as `cortex_source_scs` and/or `cortex_source_sbl`



Visualization

- ◆ NIST uses Showmesh for visualization of potentials on the cortical surface.
- ◆ Showmesh loads a mesh in .smf format,
- ◆ Loads potential distribution.
- ◆ There are options to load a point set, zoom in, out, rotate, take snapshots.

SHOWMESH TUTORIAL

NFT Matlab Scripts

- ◆ Test_script.m
- ◆ Start EEGLAB and set your parameters:
eeglab
EEG = pop_loadset('filename',eeg_file,'filepath',eeg_path);
[ALLEEG, EEG, CURRENTSET] = eeg_store(ALLEEG, EEG, 0);
% set 'of' (output folder), subject_name,
% session_name, and elec_file
subject_name = 'SubjectA';
session_name = 's1';
nl = 4; % number of layers
plotting = 1;
comp_index = 1:20; % component index for source localization

NFT Matlab Scripts

◆ Realistic modeling from MRI

% Do segmentation using the GUI

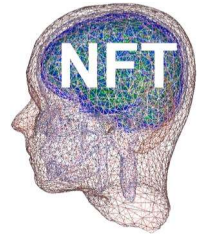
```
nft_mesh_generation(subject_name, of, nl)
```

```
nft_source_space_generation(subject_name, of)
```

% Do co-registration using the GUI

```
nft_forward_problem_solution(subject_name,  
    session_name, of);
```

```
dip1 = nft_inverse_problem_solution(subject_name,  
    session_name, of, EEG, comp_index, plotting,  
    elec_file)
```



NFT Matlab Scripts

- ◆ BEM warping mesh

```
nft_warping_mesh(subject_name, session_name,  
elec_file, nl, of, 0, 0);
```

```
nft_forward_problem_solution(subject_name,  
session_name, of);
```

```
dip1 = nft_inverse_problem_solution(subject_name,  
session_name, of, EEG, comp_index, plotting,  
elec_file)
```

NFT Matlab Scripts

- ◆ FEM warping mesh

```
session_name='s1_fem';
```

```
nft_warping_mesh(subject_name, session_name,  
elec_file, nl, of, 0, 1);
```

```
nft_fem_forward_problem_solution(subject_name,  
session_name, of);
```

```
dip1 = nft_inverse_problem_solution(subject_name,  
session_name, of, EEG, comp_index, plotting,  
elec_file)
```

NFT Matlab Scripts

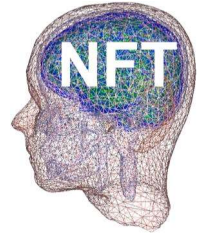
- ◆ Set NFT dipole structure to EEGLAB dipole structure

```
eeglab_folder = dirname(which('eeglab'));
```

```
mri_file = [eeglab_folder  
            '/plugins/dipfit2.2/standard_BEM/standard_mri.m  
            at'];
```

```
EEG.dipfit.mrifile = mri_file;
```

```
EEG.dipfit.model = EEG.etc.nft.model;
```

Distributed source localization

% generation of forward model:

```
nft_dsl_forward_model_generation(subject_name, of,  
mri_file)
```

% BEM forward problem solution

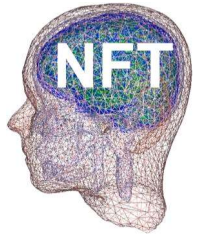
```
nft_forward_problem_solution(subject_name,  
session_name, of, 'ss_name', [subject_name 'FS_ss.dip'])
```

%FEM forward problem solution

```
nft_fem_forward_problem_solution(subject_name,  
session_name, of, 'ss_name', [subject_name 'FS_ss.dip'])
```

%cortical source localization

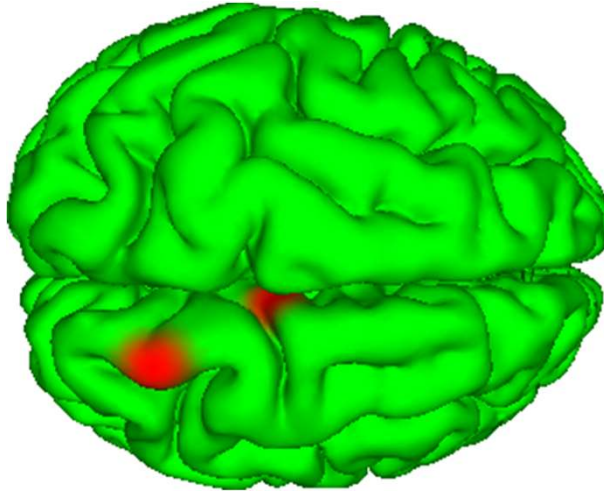
```
nft_dsl_inverse_problem_solution(subject_name,  
session_name, of, EEG, comp_index, selection)
```



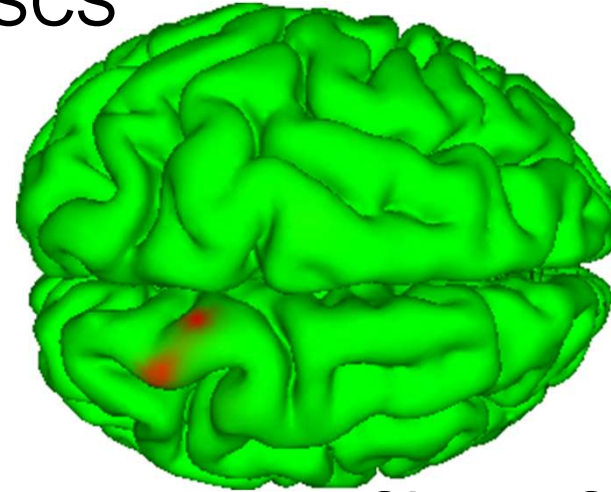
EXAMPLES

Source localization results

Simulated source

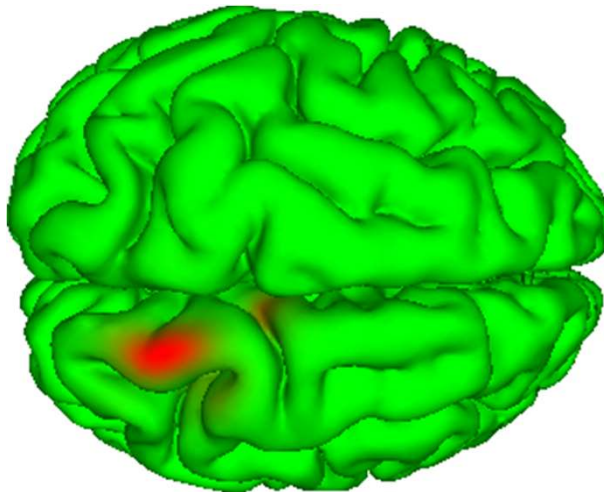


SCS

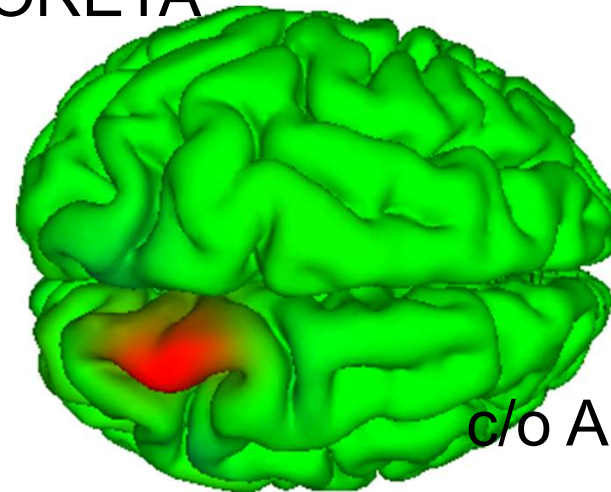


Cheng Cao, 2012

Patch-based SBL

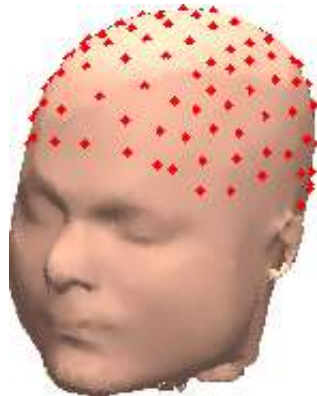


sLORETA



c/o A. Ojeda

EEG source localization



scalp



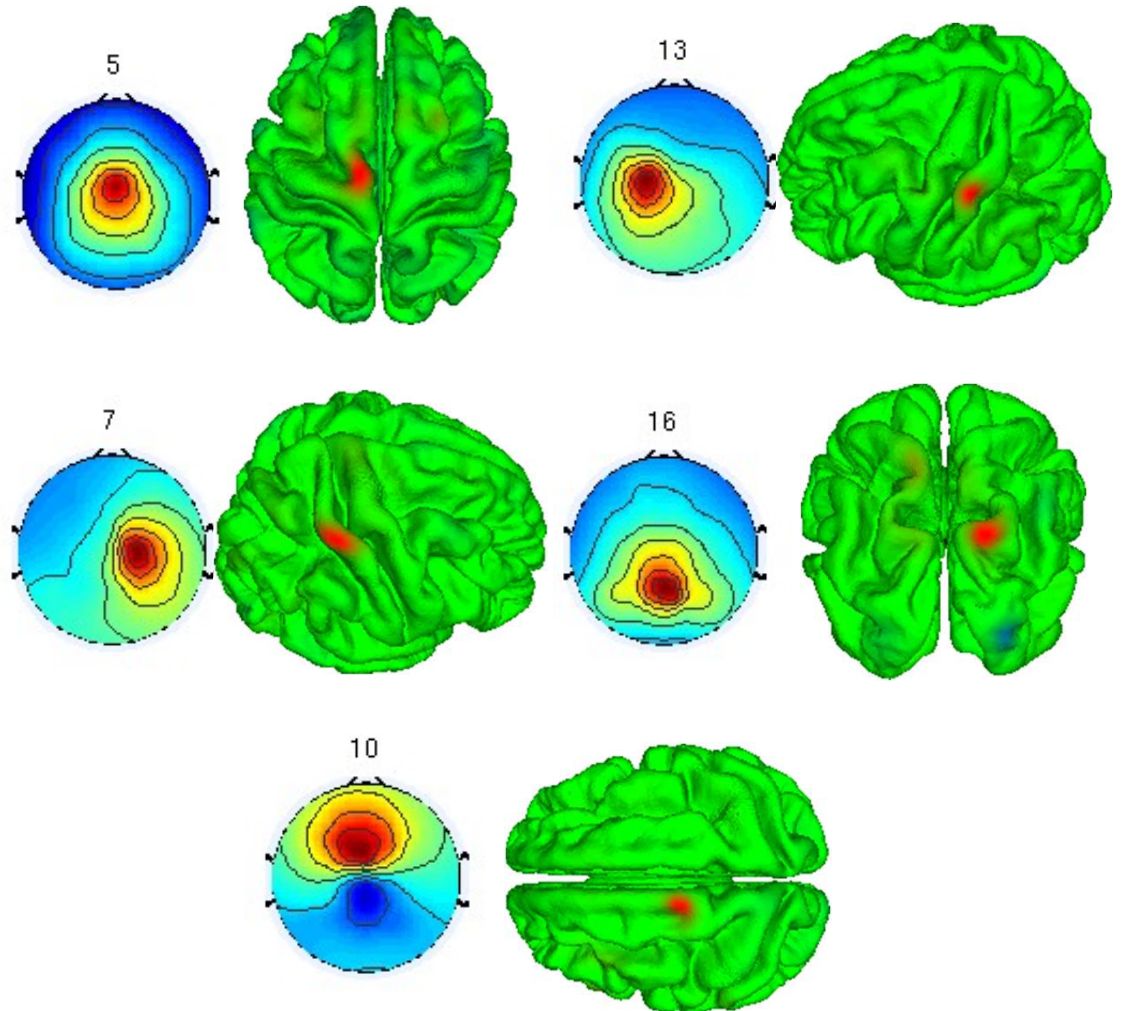
skull

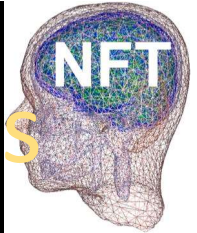


CSF



brain





Generation of individual head models

NFT (sccn.ucsd.edu/nft/) was used to generate four-layer Finite Element (FEM) head models.

6 months

12 months



1,441,777

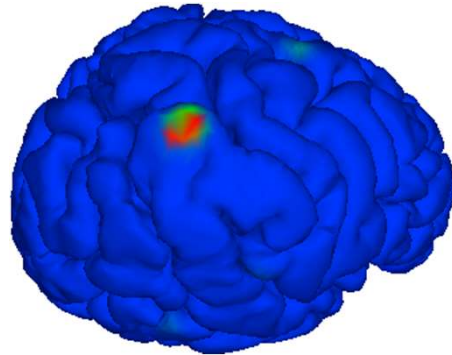
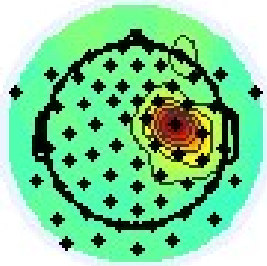
tetrahedral volume
elements

1,025,643

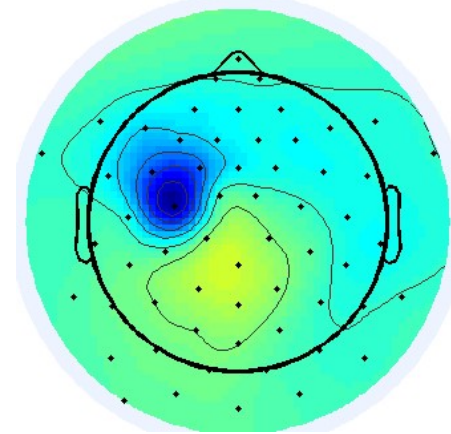
tetrahedral volume
elements

Source localization for 1-year old

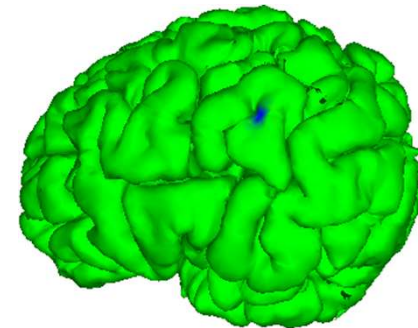
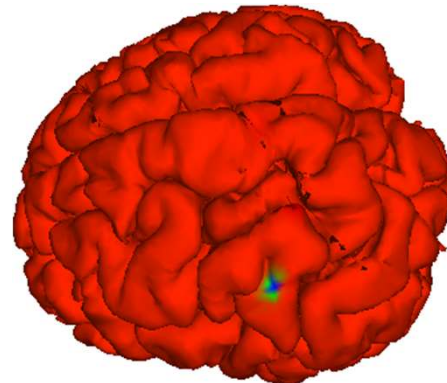
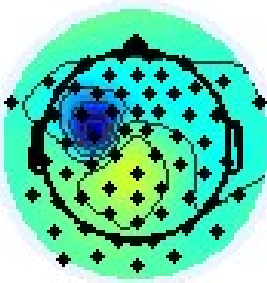
44



IC 22 from SFIC R136.12 epochs

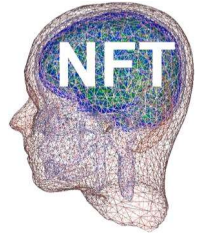


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Future work

- ◆ Neuroscience Gateway Portal (NSG)
 - Dipole source localization
 - Distributed source localization
- ◆ SCALE
 - Simultaneous skull conductivity and source localization estimation.
 - Inputs: EEG and MRI
 - Outputs: Source estimates and skull conductivity estimate.



NFT download

NFT wiki

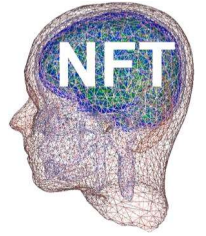
- ◆ <https://github.com/sccn/NFT/wiki>

NFT code

- ◆ <https://github.com/sccn/NFT>

NFT test data

- ◆ https://github.com/sccn/NFT_test



NFT reference

- ◆ Akalin Acar Z, Makeig S, Neuroelectromagnetic Forward Head Modeling Toolbox, J. of Neuroscience Methods, vol 190(2), 258-270, 2010.